

Introduction to Polymer Materials

Lecture 11: Polymer Coatings

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Polymeric Coatings

Coatings are materials that are applied to a *surface* which form a *continuous film* in order to *beautify* and/or *protect* the surface.

- Sealants
- Paint
- Enamel
- Varnish
- Shellacks
- Stains



Coating Market

50+ billion USD worldwide, divided into 3 main segments

Architectural: Paints, varnishes, and lacquers for direct application to interior or exterior surfaces of buildings

- ~50% of total market, but lowest profit margin
- Generally air-dried
- Sherwin-Williams, Benjamin Moore, ICI Paints

OEM/Product: Applied to equipment in a manufacture process

- Appliances, cars, industrial machinery, furniture, ...
- ~35% of total market, higher profits
- Baked, radiation-cured, electrostatic-spray
- Automotive: PPG, DuPont, BASF

Specialty Market: Everything else

- Auto refinish, traffic marking, ...
- ~15%, usually high-value
- Air or force dried
- PPG, DuPont, Akzo Nobel, ...

OEM = original equipment manufacturer

Enamels

- Not the same as vitreous or porcelain enamel-a glass powder fired in a furnace
- In paint, it refers to a higher quality paint



Sealants

- Barrier material (protective coating)
- Generally weaker than adhesive

examples:

- Asphalt/coal tar based seal coatings

- Anerobic acrylic sealants

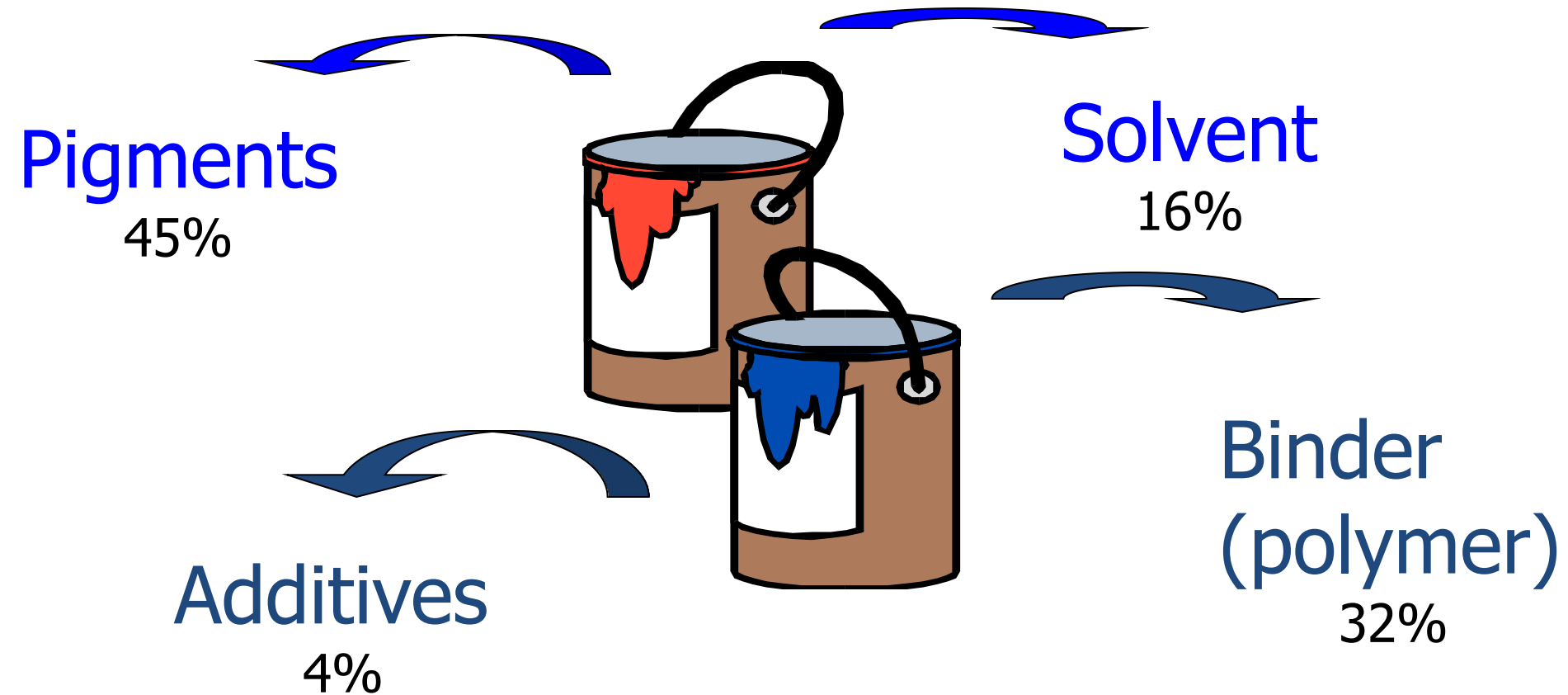
- silicone coatings on membranes

- roof sealants

What are roofing sealants? Filled and highly pigmented elastomers

- EPDM (ethylene-propylene-diene monomer) rubber (e.g. liquid rubber)
- Acrylic elastomers
- Silicones (not very good-often adhesion issues)
- Polyurethane

What is in a can of paint?

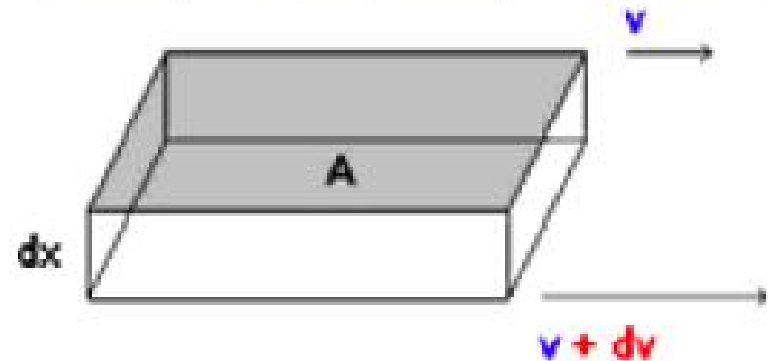


Water based paints may have considerably more liquid carrier

1 pascal second = 10 poise = 1,000 millipascal second
1 centipoise = 1 millipascal second

Viscosity

- ◆ Fluids resist a gradient of flow
 - Tries to achieve constant speed across the sample
- ◆ This resistance creates a force called the **viscous drag**
- ◆ $\eta \equiv$ viscosity coefficient (or just viscosity)
- ◆ Viscosity is measured in **poise** ($10^{-1} \text{ kg m}^{-1} \text{ s}^{-1}$)



$$F_{drag} = \eta A \frac{dv}{dx}$$

High viscosity coating



Measuring viscosities



10-100,000 cP



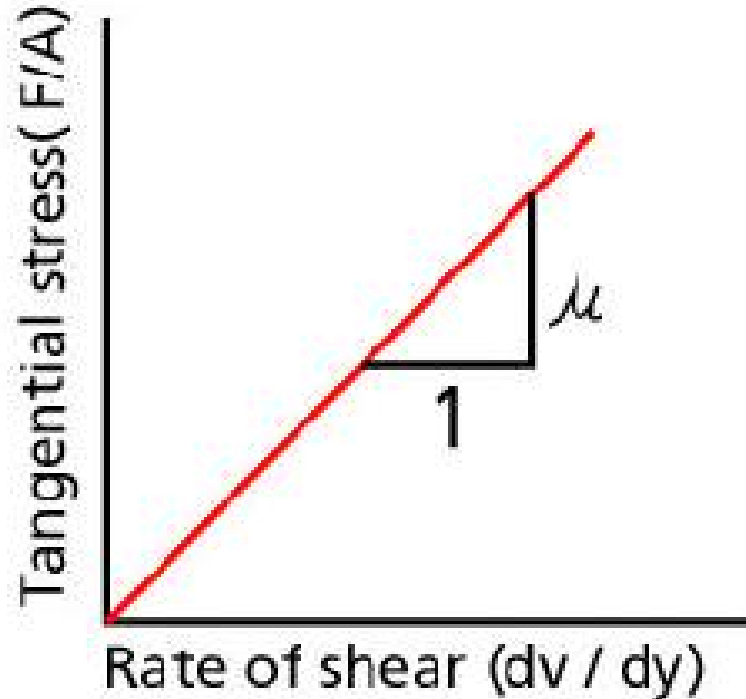
Requires standards

Measuring viscosities

Zahn Cup



Newtonian Fluids



Newtonian flow paints would also flow away from sharp corners and edges and give significant sagging or curtaining problems when applied.

MATERIAL

APPROXIMATE VISCOSITY

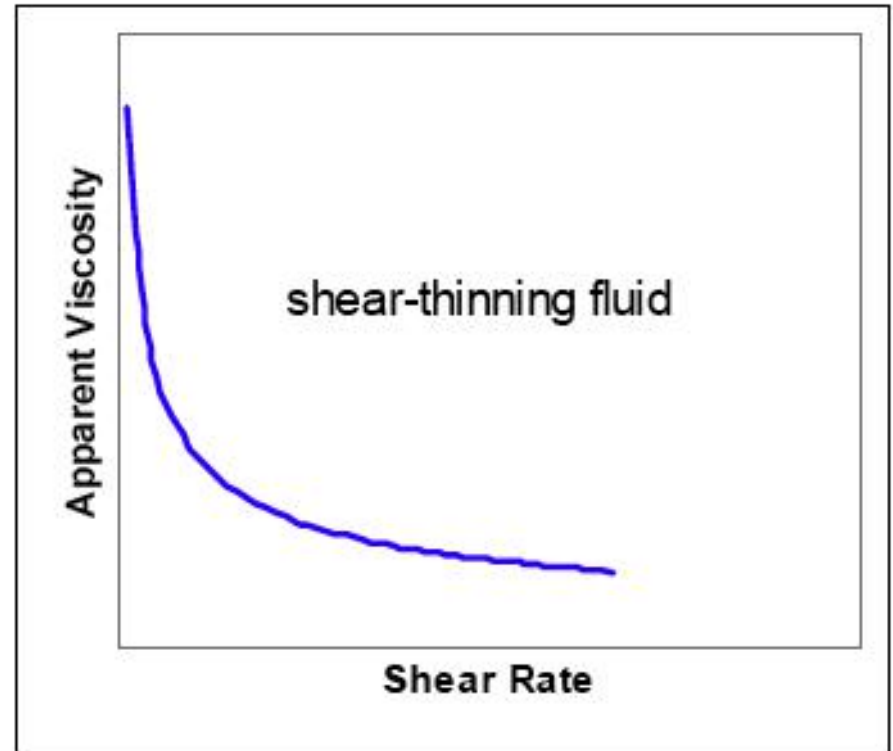
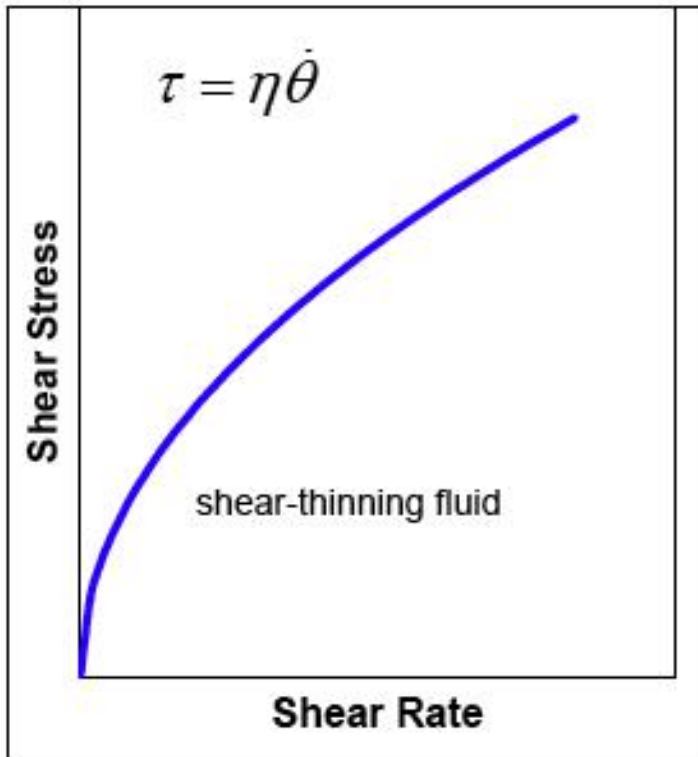
(in centipoise)

Water @ 70 F	1 to 5
Blood or Kerosene	10
Anti-Freeze or Ethylene Glycol	15
Motor Oil SAE10 or Mazola Corn Oil	50 to 100
Motor Oil SAE30 or Maple Syrup	150 to 200
Motor Oil SAE40 or Castor Oil	250 to 500
Motor Oil SAE60 or Glycerin	1,000 to 2,000
Karo Corn Syrup or Honey	2,000 to 3,000
Blackstrap Molasses	5,000 to 10,000
Hershey Chocolate Syrup	10,000 to 25,000
Heinz Ketchup or French's Mustard	50,000 to 70,000
Tomato Paste or Peanut Butter	150,000 to 250,000
Crisco Shortening or Lard	1,000,000 to 2,000,000
Caulking Compound	5,000,000 to 10,000,000
Window Putty	100,000,000

Non-Newtonian Liquids



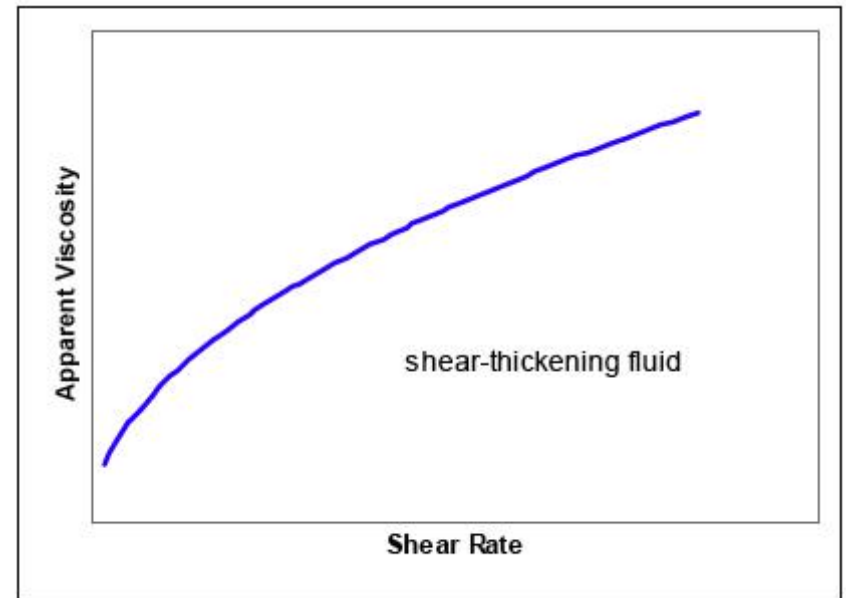
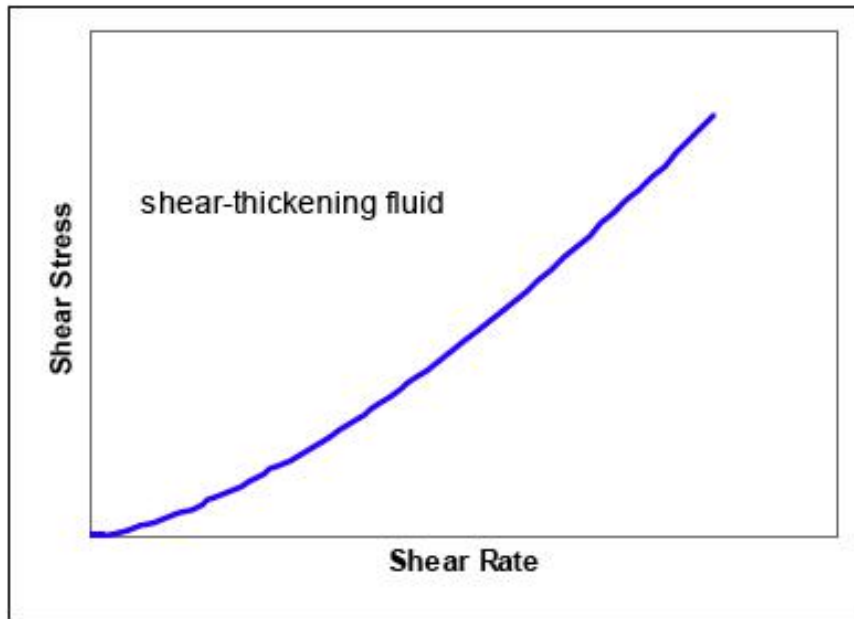
Non-Newtonian Liquids: (Time Independent) Pseudo-Plastic



- molten thermoplastics
- polymer solutions such as polyethylene oxide in water
- some paints.
- shampoo
- ketchup

Shear Thickening (Time Independent) Dilatant

Solids dispersed in liquid



Corn starch, clay slurries, wet sand (quick sand)
surfactant solutions, peanut butter, filled composites

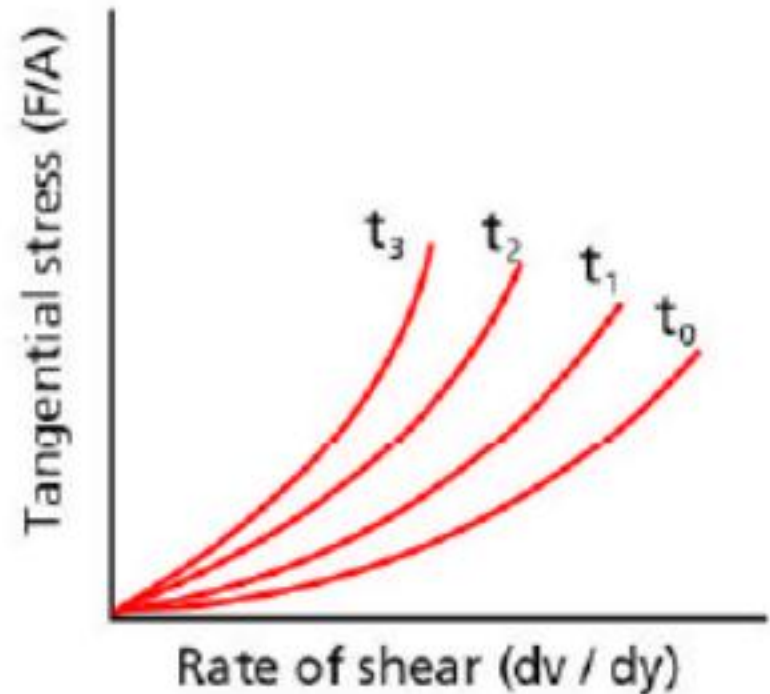
Time dependent Non-Newtonian Viscosity

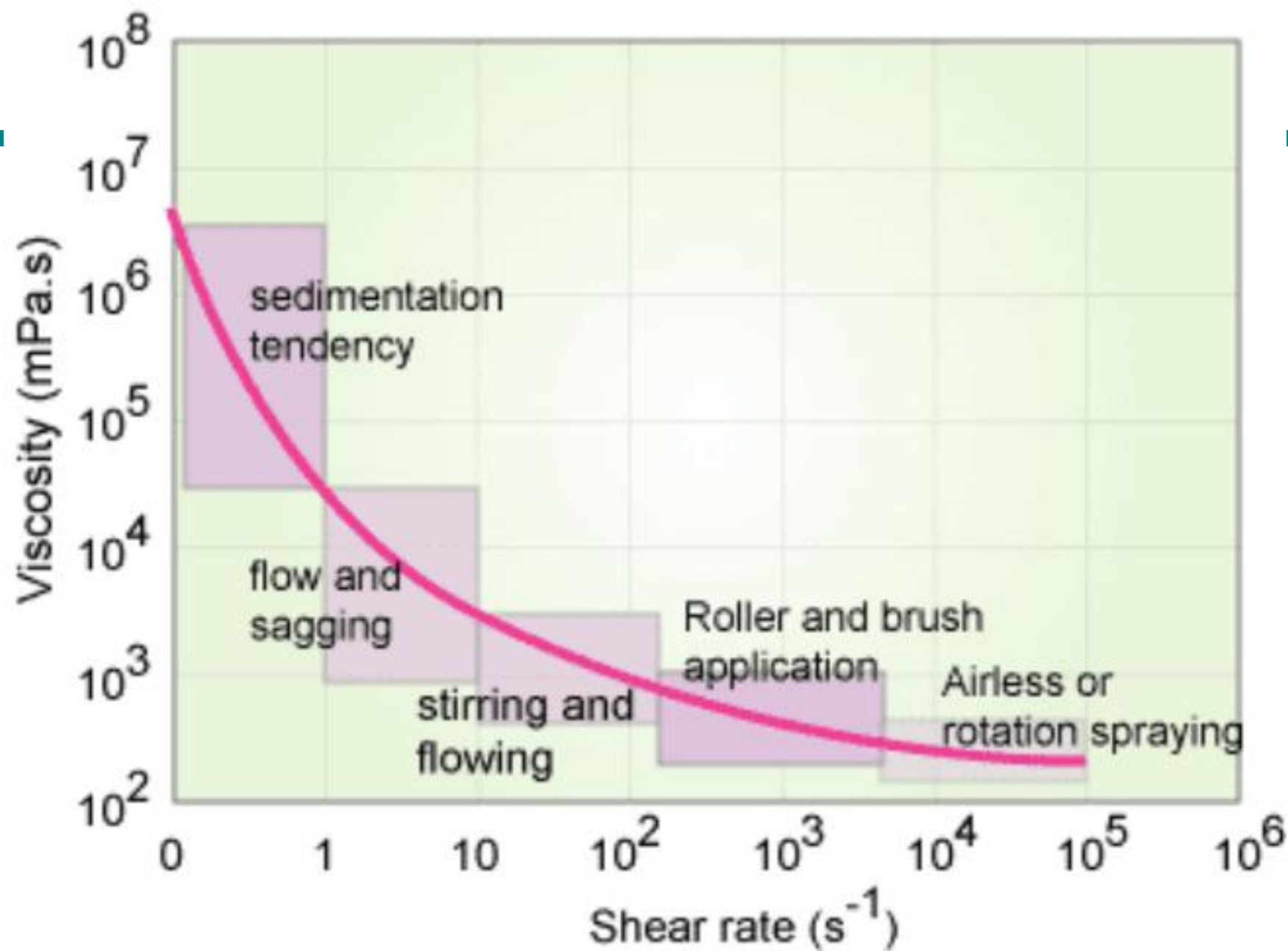
The viscosity of the fluid is dependent on temperature, shear rate and time.

Depending on how viscosity changes with time the flow behavior is characterized as:

- Thixotropic (time thinning, i.e. viscosity decreases with time) yoghurt, paint
- Rheopetic (time thickening, i.e. viscosity increases with time) gypsum paste

Thixotropic fluids are quite common in chemical as well as in food industry.
Rheopetic fluids are very rare.



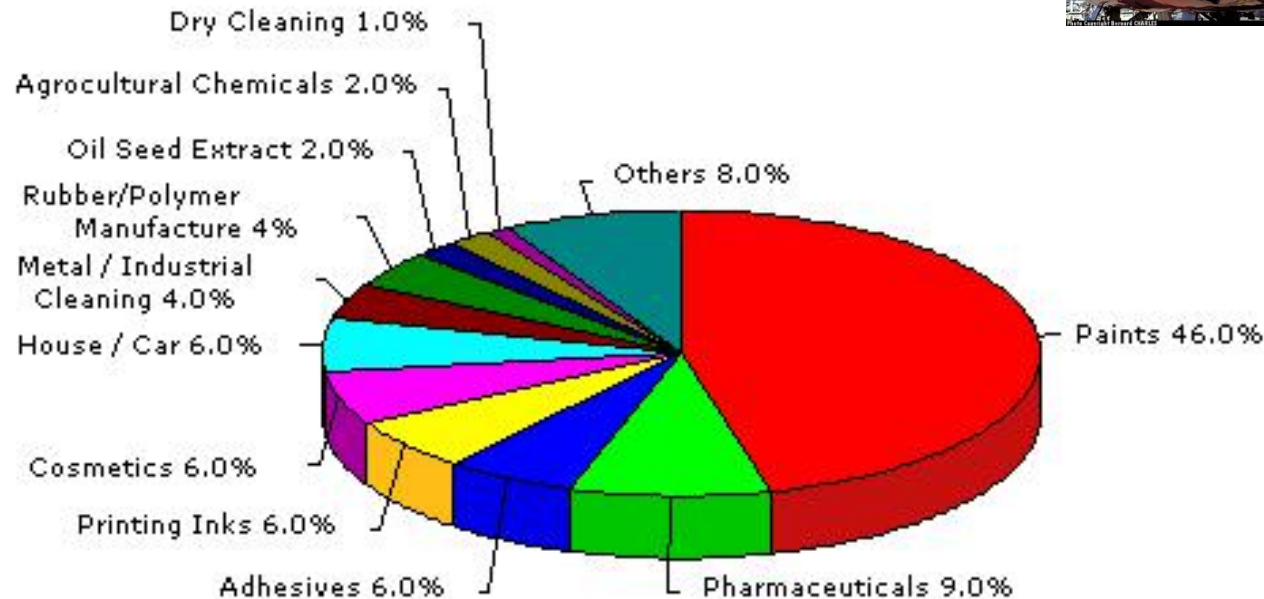


Paint: Interesting Facts

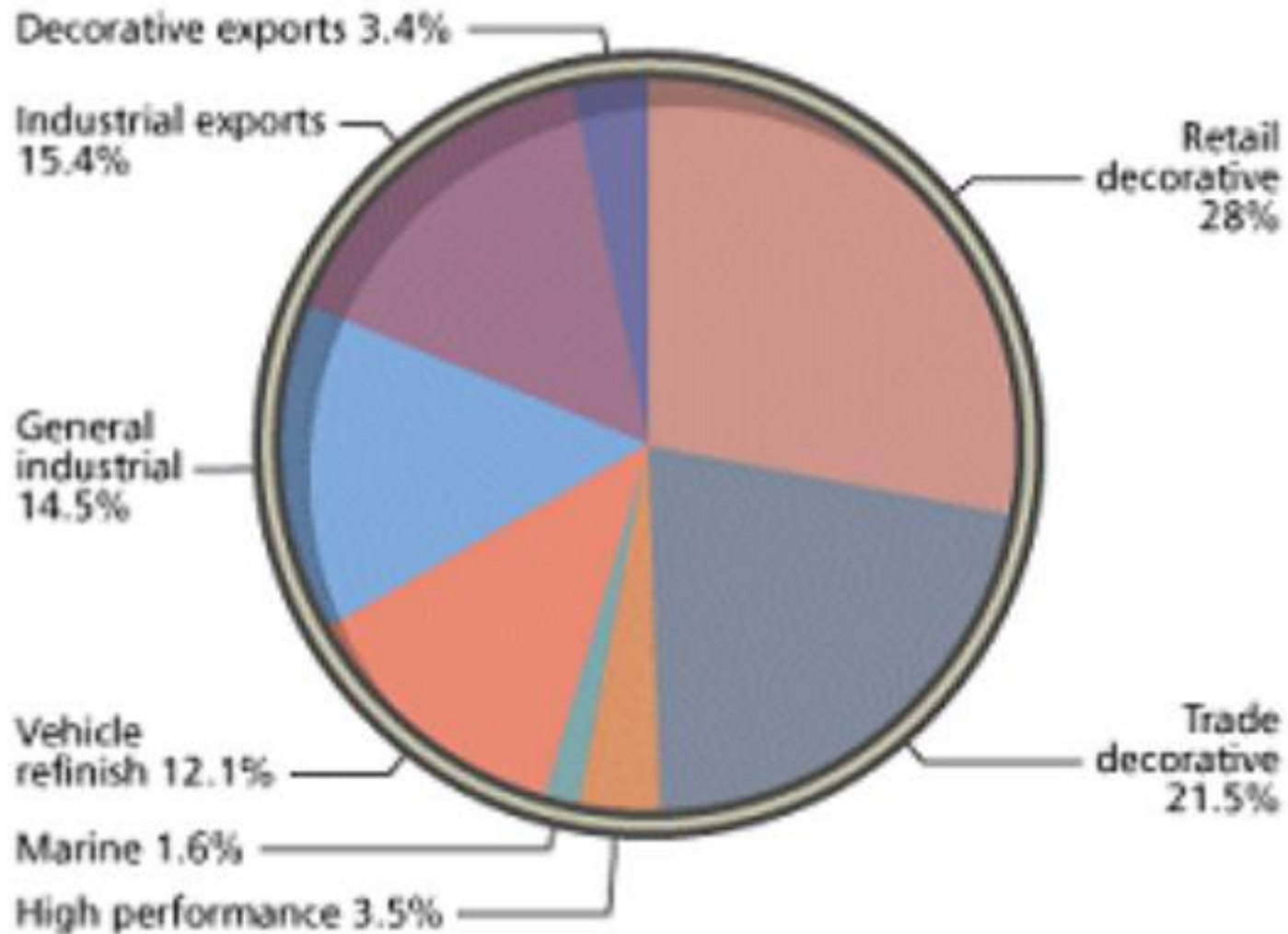
A jumbo jet needs 2 tons of paint.
The world's shipping fleet would produce an extra 70 million tons of greenhouse gasses and nearly 6 million tons of acid-rain-producing sulfur dioxide if ships were not treated with anti-fouling paints



SOLVENT USES



Paint Markets



Contents of Paint

- **Pigment**
 - Provides color and durability
 - Also improves the strength of the paint
- **Binder**
 - Holds the pigment in liquid form
 - When applied it then gives the paint the ability to adhere to the surface.
- **Solvent**
 - Effectively thins the paint
 - It carries the pigment and binder
 - Used to regulate how much a paint flows
 - Called a “thinner” when used with lacquer
 - Called a “reducer” when used with enamel

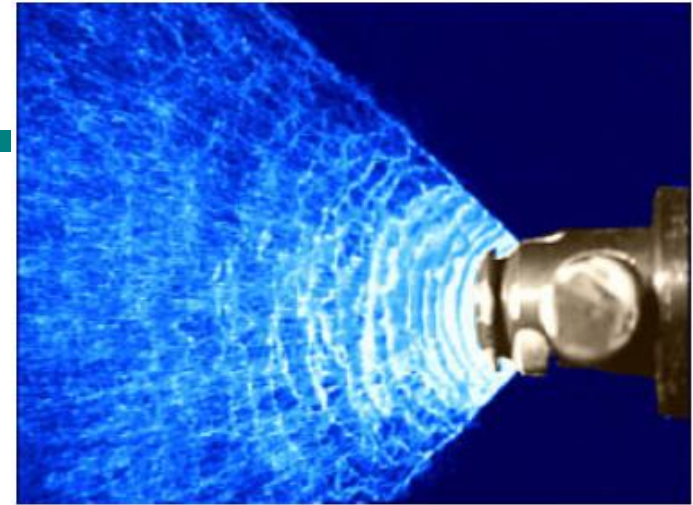


Figure 2. Microsecond y-z-plane image of latex paint sheet atomization by an airless spraygun.

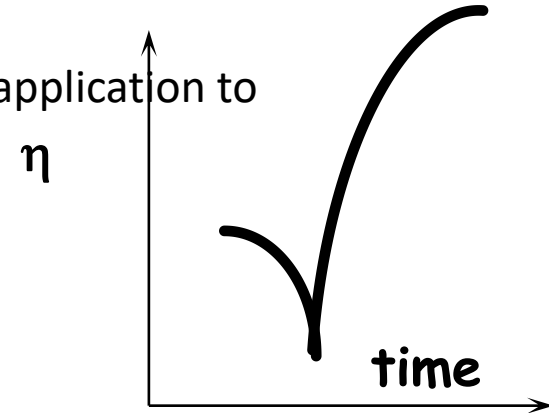


General Coating Formulations

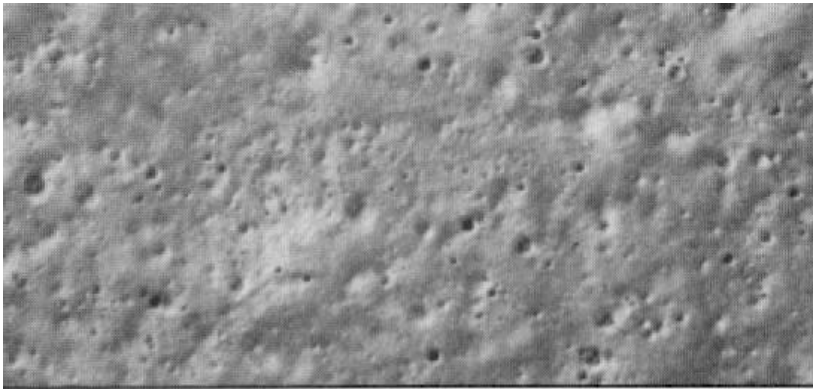
Method	Polymer M.W.	Solids	Polymer structure	Dry rate (no heat)	Min. dry temp.	Handling/ Storage	Examples
Evaporation	high	(i) low, 10-35% solution (ii) medium 40-70% emulsion	linear	fast	no practical limit (solutions)	good	nitrocellulose other lacquers; some emulsion paints
Chemical reaction between paint and air	low	medium to high, 35-100%	crosslinked	slow-moderate	very slow in cold weather	cans must be well sealed	decorative paints some stoving enamels;
Chemical reaction between paint ingredients	low or very low	medium to high 30-100%	crosslinked	fairly fast	varies; 10-15° C	two-pack or short shelf life, unless stoving or radiation curing type	industrial stoving finishes; acid-catalysed polyurethane and polyester wood finishes

Desired Rheology of Coatings

- Coating viscosity requirements change on passing from the application to drying stages of the process.
- Application
 - brushing, rolling, spraying processes generate high shear, and favour low viscosity formulations
 - chain orientation and surface imperfections (brush strokes) must be relieved by flow.
- Film Formation - Drying
 - sagging during the drying process must be countered by low-shear viscosity.
- Final Coating Properties
 - permanent coating requires near infinite viscosity, through high molecular weight (lacquers) or crosslinking (enamels).



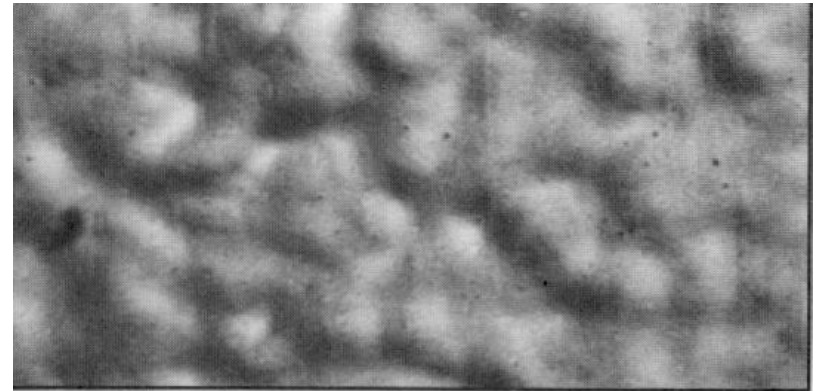
Solvent-Related Surface Defects



Trapped bubbles in an industrial paint caused by air inclusion and/or solvent popping (100X)

Orange peel patterns result from high film viscosity and/or surface tension gradient induced flow.

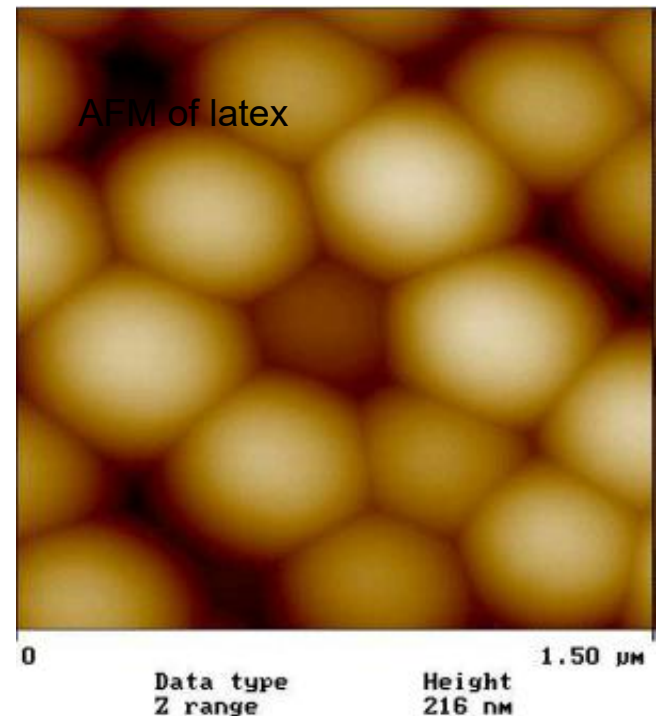
Solvent popping results from rapid solvent evaporation and an inability of the coating to flow.



A typical orange peel pattern in a baked industrial enamel (15X)

Binders: Polymers

- Thermosets
 - Alkyds
 - Epoxies
 - Urethanes
 - Formaldehyde resins
 - Alkyds (classic oil paint)
- Thermoplastics (latexes)
 - Acrylics + other vinyl monomers
 - Vinyls: vinylacetate copolymers with acrylics, vinyl chloride, styrene



Binder: alkyd resin

Polyester of:

- Polyol (glycerol)
- Phthalic acid
- Fatty acid



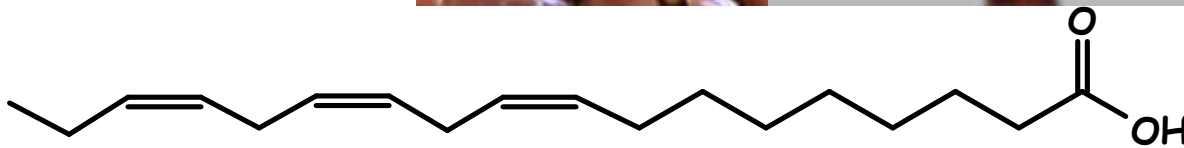
Linseed



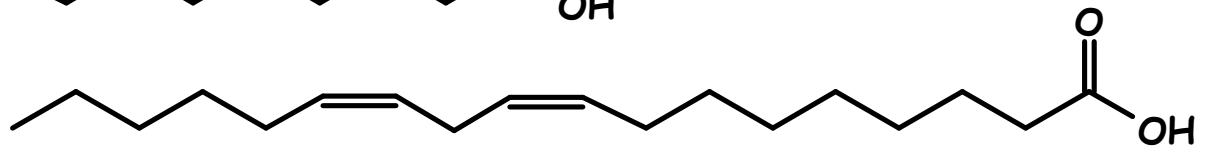
Alkyd resin



Sunflower



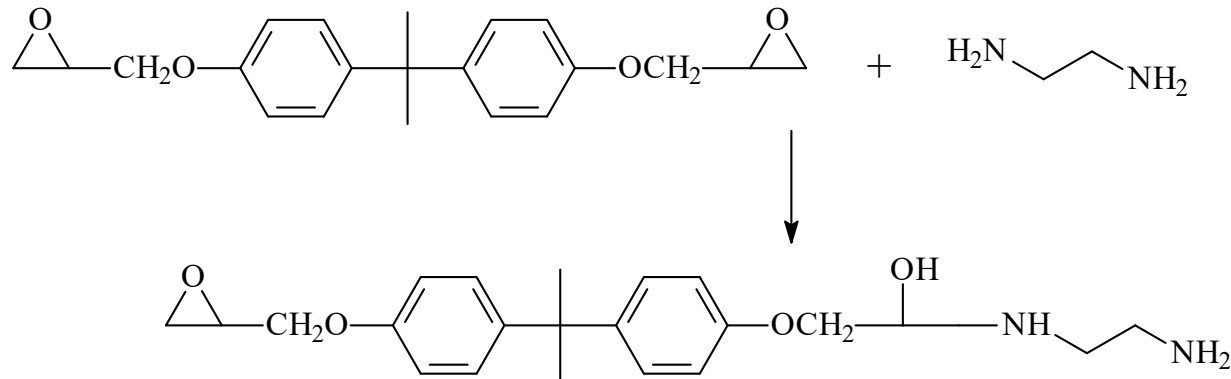
Linolenic acid



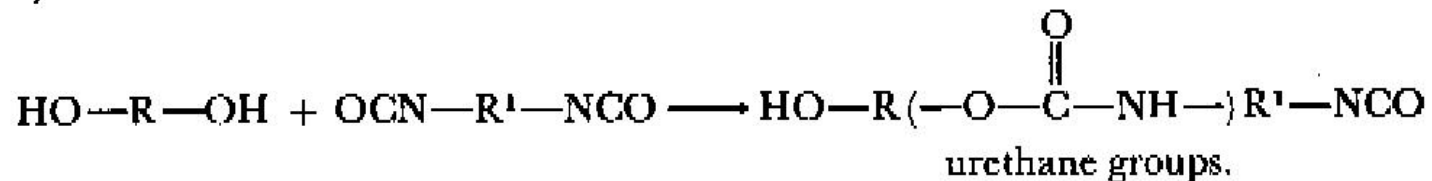
Linoleic acid

Thermosetting Binders: Epoxy and Polyurethane Resins

- Epoxy resins are two-component paints formulated from epoxide functionalized monomer and (usually) amine hardeners.



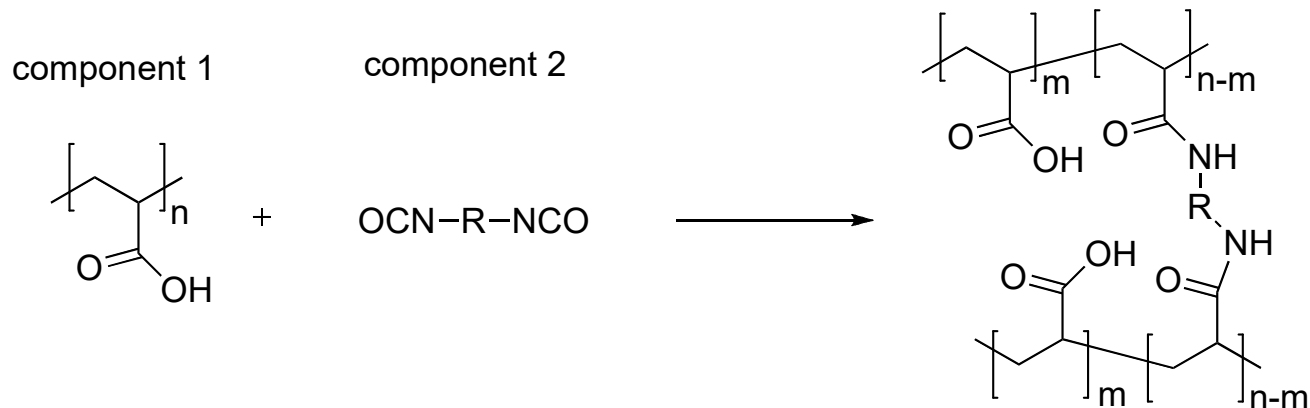
- Reaction of diisocyanates with diols generates polyurethane coatings whose structure/properties can be varied widely.

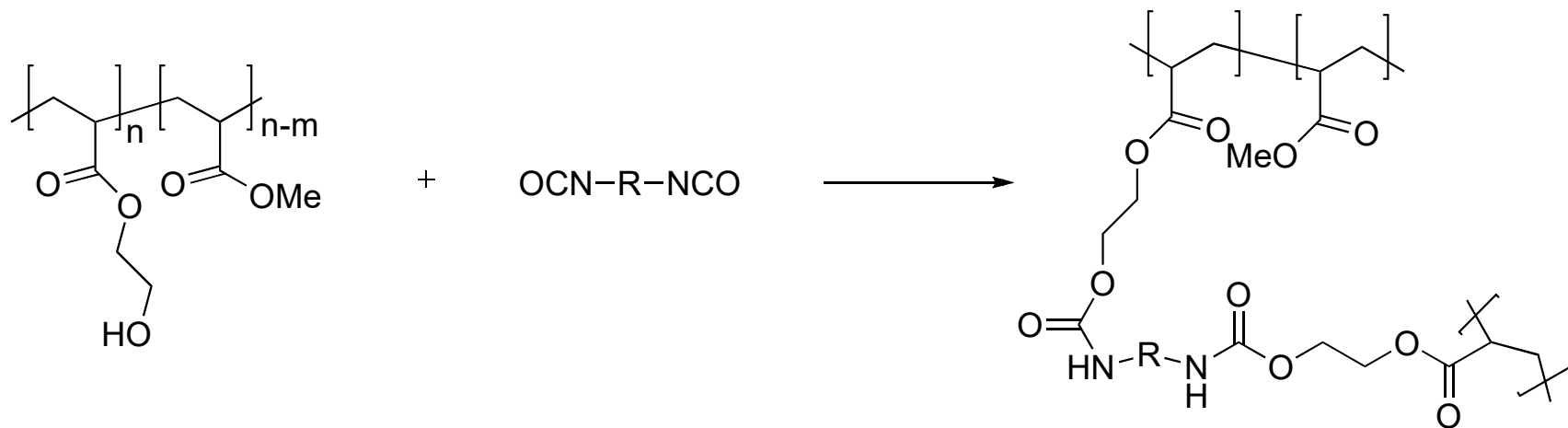


- Polyurethanes afford superior abrasion and chemical resistance, as well as a fast, low-temperature cure.

Polyurethanes

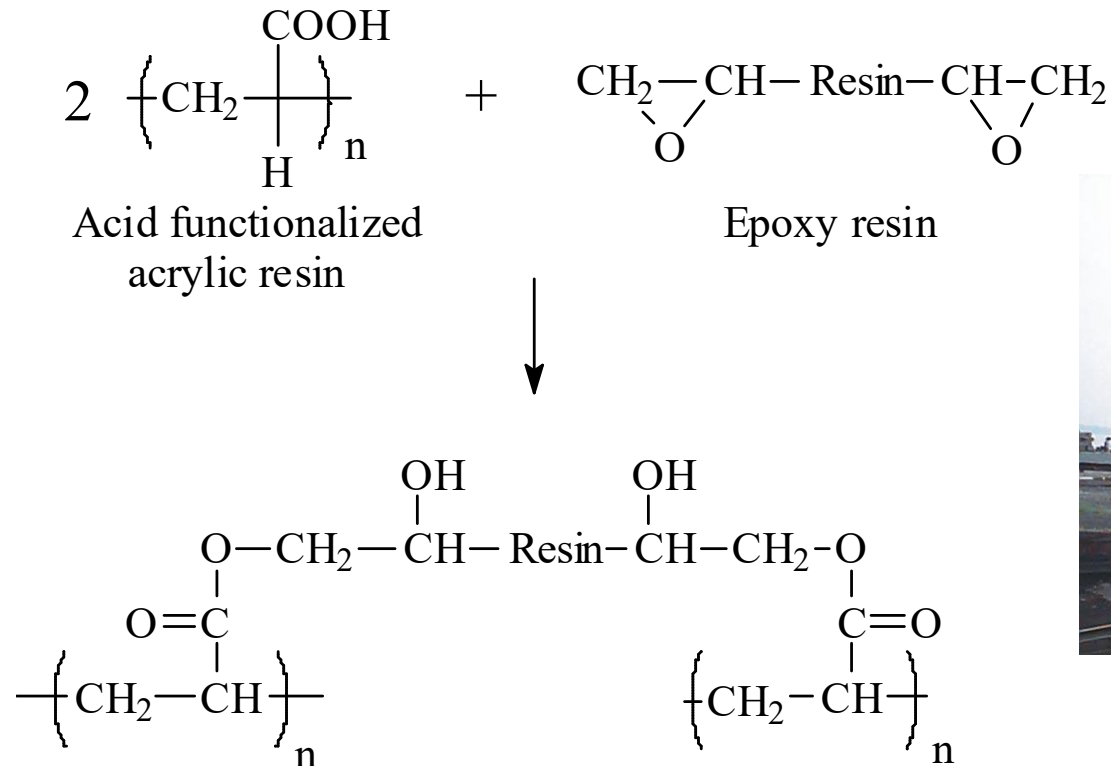
- One part polyurethane
- Moisture cured polyurethane
- Acrylic polyols-aliphatic linear isocyanate two part polyurethanes
- Polyester polyols-aliphatic isocyanate two part polyurethane



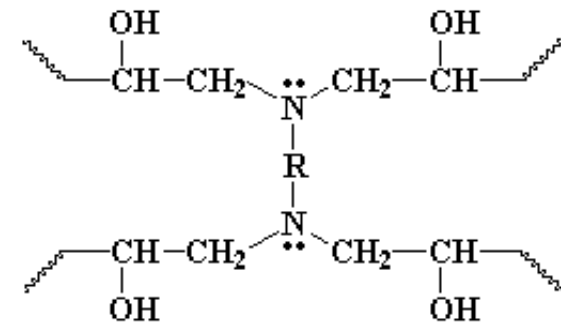
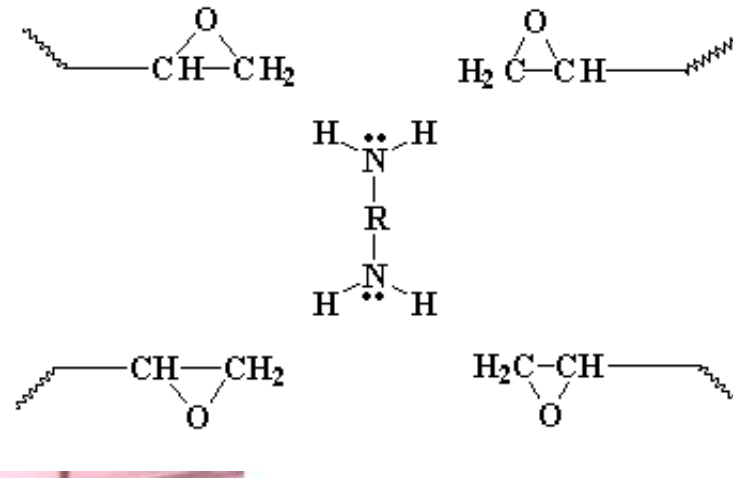


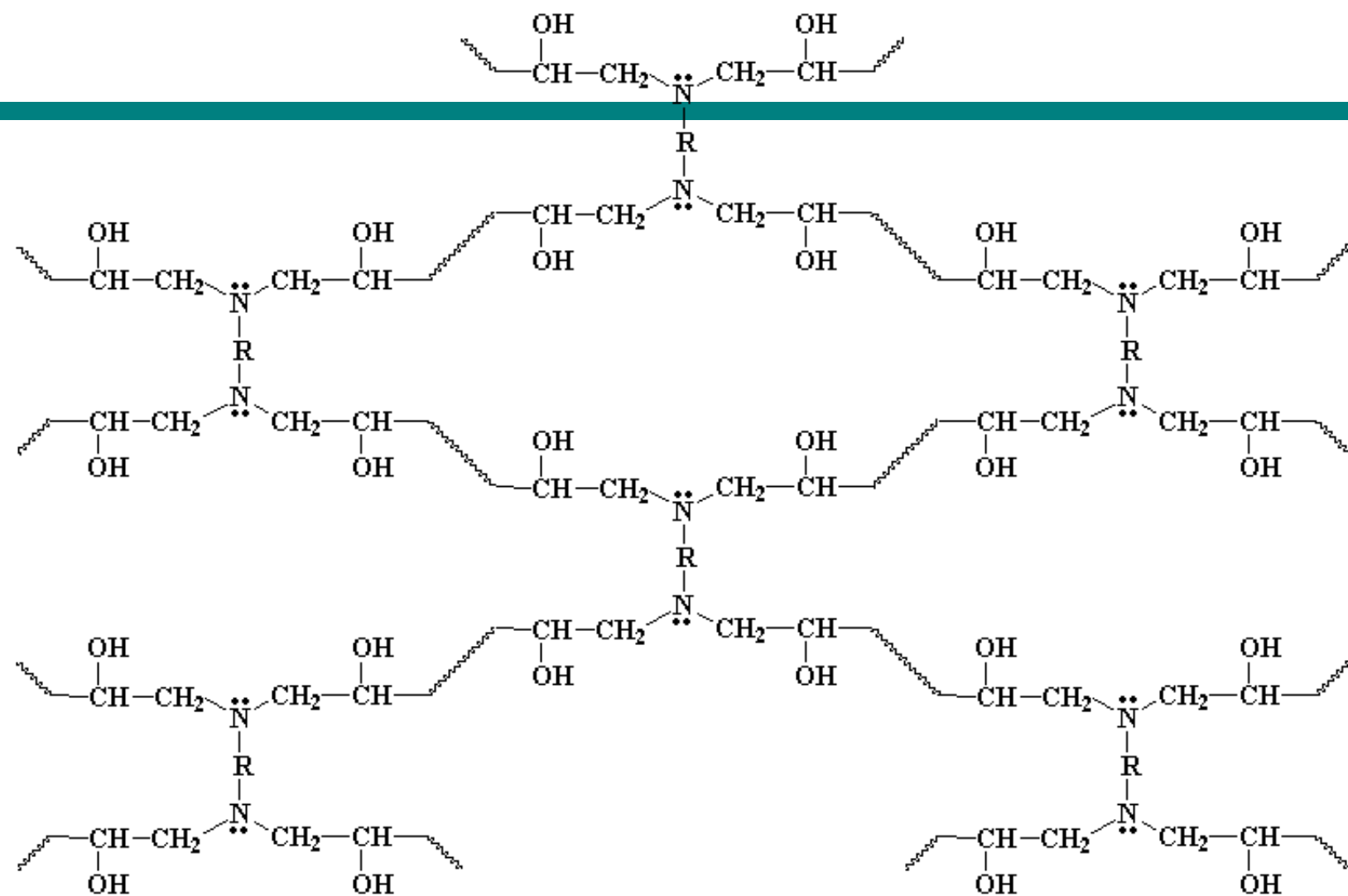
Thermosetting Binders: Combination

- Copolymerization of an acid-functionalized acrylic resin and an epoxy resin yields a crosslinked, block copolymer coating.



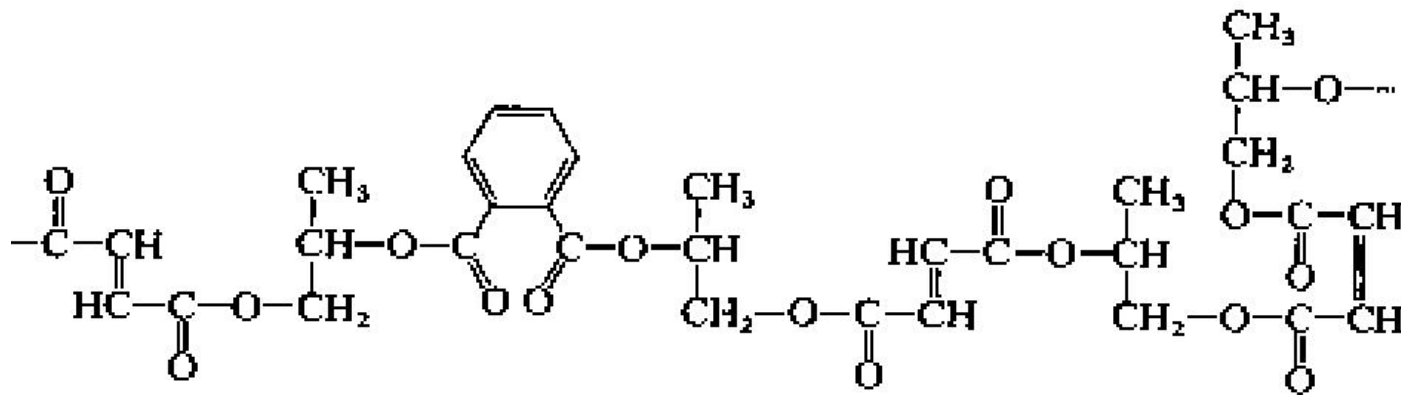
Epoxy Paints





Thermosetting Binders: Cured Polyesters and Acrylics

- Unsaturated polyesters and acrylic resins of low molecular weight can be polymerized by free radical addition chemistry to generate a stable, crosslinked film.
- Consider a resin comprised of 1,2-propylene glycol, phthalic anhydride and maleic anhydride.



- Free radical polymerization initiated by an organic peroxide generates a networked structure of high molecular weight by addition through unsaturation in the polymer backbone.

Thermosetting Binders: Oxidative Drying Alkyds

- While alkyds can be classified as polyesters, the term is reserved for oil-based finishes.
- Oils are first transformed into monoglycerides:

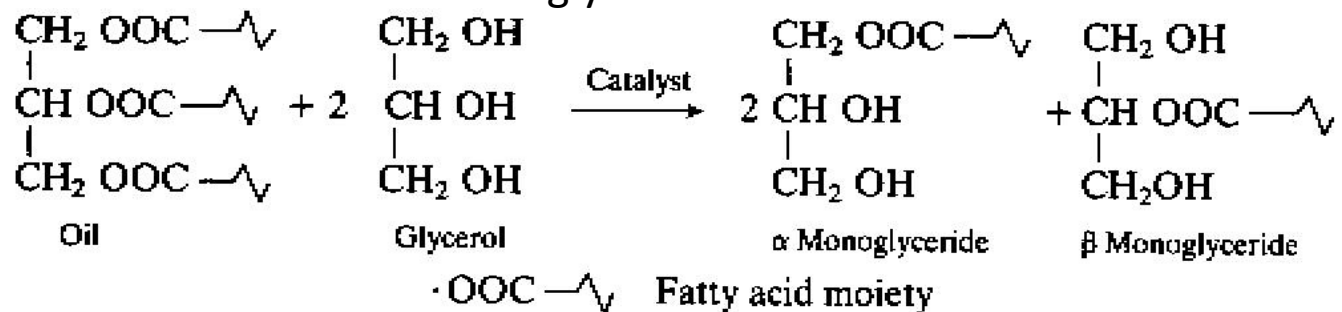
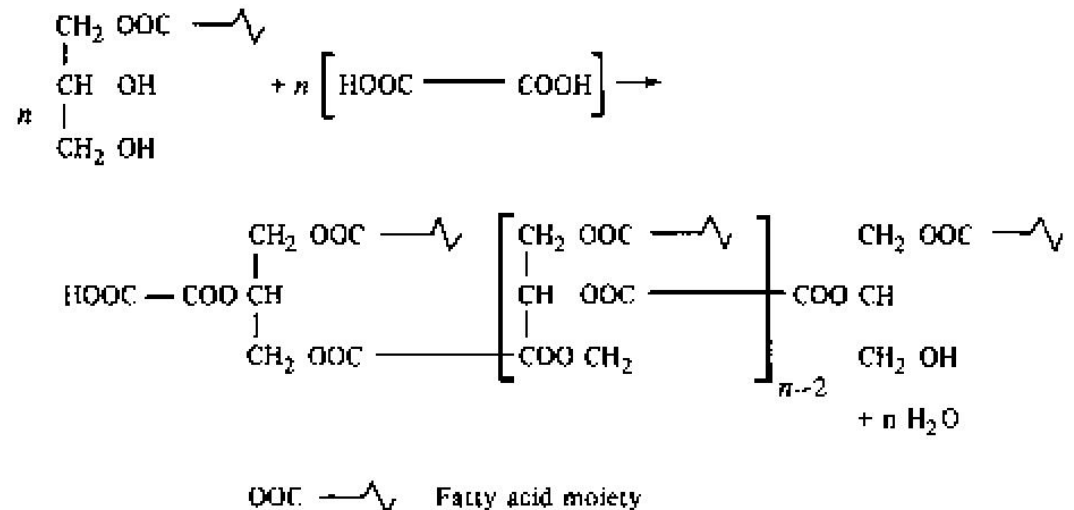


Fig. 2.5 — Schematic representation of monoglyceride formation.

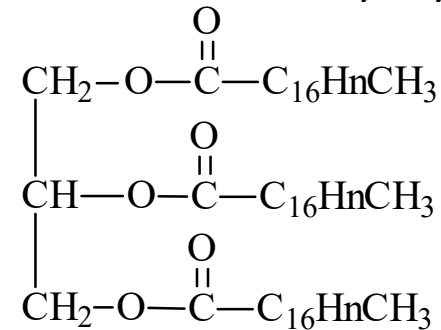
- Film formation results from condensation polymerization with diacid as well as oxidative cure.



Thermosetting Binders: Oxidative Drying Oils

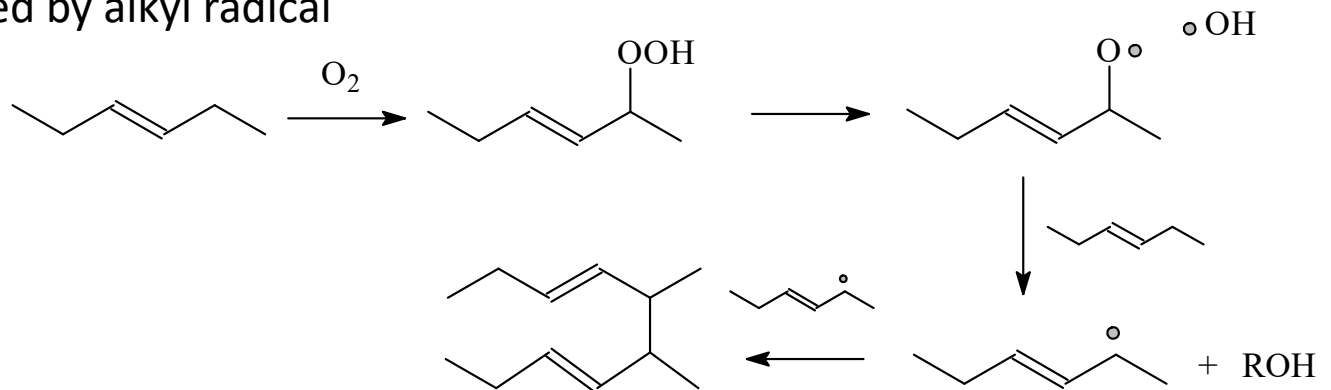
- Coatings containing oil-based films are no longer used as finishes due to poor gloss, soft films and inferior water resistance.
 - Oils are frequently used in conjunction with other resins to modify drying properties and film structure.

- Natural oils are extracted from linseed, soya bean, coconut, etc.
- Unsaturated oils are valued for their relatively rapid oxidative curing.



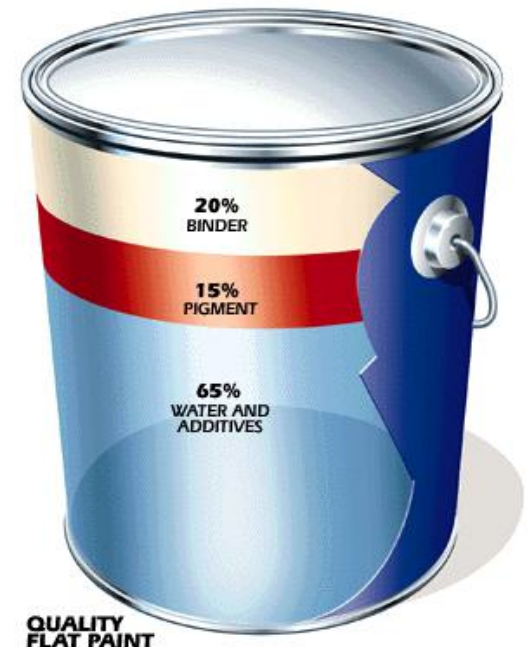
$n=32,30,28,26$

- Curing occurs through hydroperoxide formation, followed by alkyl radical combination.



Water-based formulations: Emulsions

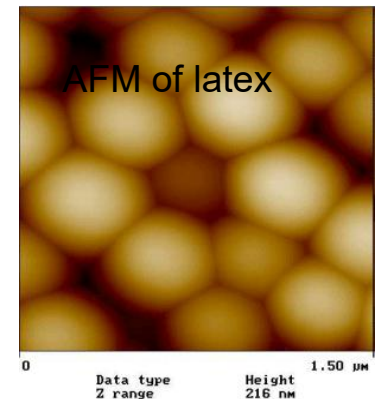
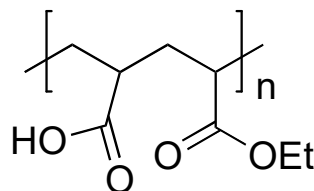
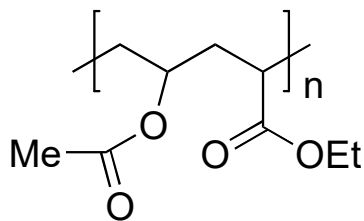
- Emulsion formulations were developed for environmental reasons and for the delivery of very high molecular weight binders.
 - Water is the continuous phase, which results in a very low viscosity coating.
 - Thixotropic agents are required to raise the zero-shear viscosity of the formulation.
- Most emulsion paints contain some solvent/plasticizer to modify the T_g of the polymer.
 - Film formation requires coalescence of polymer particles, which cannot occur below T_g .
 - Organic solvents assist with film formation, and evaporate to leave a solid coating.
 - Alternately, a plasticizing agent is used to maintain a flexible film throughout the object's lifetime.



1 A can of high-quality latex paint contains 20% binder by volume. In traditional latex paint, there is 1% to 7% solvent dissolved into the binder. This solvent is largely responsible for giving latex paint its odor. New, low-odor varieties of latex paint are essentially solvent-free, though they contain about the same amount of binder.

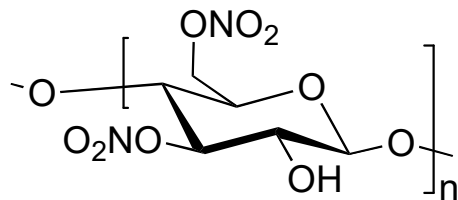
Thermoplastic Binders: Emulsions

- Household emulsion paints are usually comprised of
- poly(vinyl acetate-co-ethyl acrylate) or poly(acrylate-co-acrylic acid) resins
 - pigment is dispersed in the continuous aqueous phase with suitable surfactants and water-soluble thickener.
 - plasticizers or volatile solvents are used to lower T_g such that particle coalescence can function
 - High-gloss latex paints cannot be manufactured, as surface uniformity is generally poor
 - Residual surfactant can lead to inferior water stability of latex derived films.



Thermoplastic Binders: Lacquers

- Lacquers harden quickly at all practical temperatures, are supplied in one pack and do not suffer from shelf or pot life problems.
 - comprised of **hard** linear polymers in solution
- Cellulose nitrate, a derivative of the natural product cellulose is prepared with varying degrees of modification for different grades:



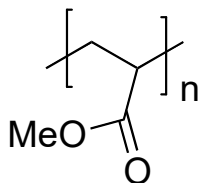
Solubility in esters, ketones

and alcohols depends on

extent of cellulose

functionalization

- Acrylic lacquers are comprised of homo or copolymers of acrylates, properties depending on polymer composition distribution:



Poly(methyl methacrylate)

provides hardness and UV stability. Plasticizers and

copolymerization alters T_g.

Coating Formulations: Solvent Selection Criteria

- Solvating Capacity:

Miscibility of polymer/solvent systems are dictated by thermodynamics, as approximated by solubility parameters and hydrogen bonding groupings.
- Viscosity:

Influenced by solvating capacity, but also a function of the viscosity of pure solvent and additives.
- Volatility:

Rate of solvent evaporation influences drying time as well as film aesthetic qualities. Decisions often based upon boiling point/range.
- Toxicity and smell.
- Cost.

Coating Formulations: Extenders and others

- Extenders provide no colour to a film, but their use is an inexpensive method of improving adhesion, ease of sanding, film strength and opacity.
 - Calcium carbonate (whitewash)
 - Aluminum silicate (clay)
 - Magnesium silicate (talc)
 - Barium sulphate (barytes)
 - Silica
- Viscosity Modifiers
 - silicates, clays, poorly soluble resins
- Dispersion Aids
 - aid in pigment dispersion - chosen on a case-by-case basis
- Interfacial Tension Modifiers
 - non-ionic surfactants, soaps
- Biocides
 - insecticides, fungicides

Coating Formulations: Pigments

Pigments are selected on the basis of:

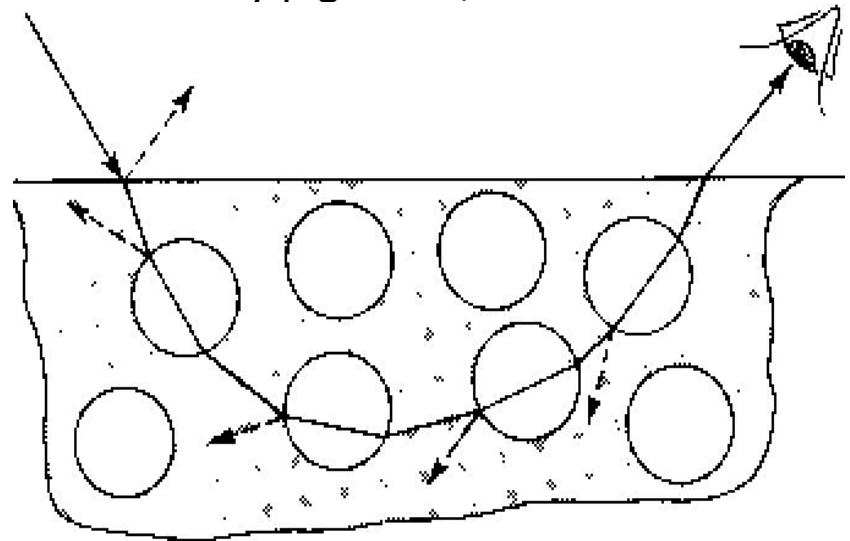
Particle size	Particle shape	Refractive Index
Tinting strength	Lightfastness	Hiding Power
Thermal Stability	Chemical Reactivity	Density (cost)

Property	Preference	Reasons
(1) Brilliance and clarity of hue	Organic	The most attractive, cleanest colours are obtained with organic pigments.
(2) White and black paints	Inorganic	The purest white pigment is TiO_2 and the most jet black, carbon.
(3) Non-bleeding	Inorganic	Inorganic compounds have negligible solubilities in organic solvents. Some organics are very insoluble.
(4) Lightfastness	Inorganic	Inorganic compounds are generally more stable to UV than organics.
(5) Heat stability	Inorganic	Very few organic compounds are stable above 300°C .



Aesthetic Properties of Dried Film Coatings

- Opacity
 - Extent of substrate coverage, as determined by pigments, extenders and other occlusions in the film.
 - Dependent on refractive index of fillers relative to the polymeric binder.
- Surface Finish
 - Gloss is a function of surface irregularity, as determined by the film formation process and dispersion of pigments/fillers.
- Color
 - Inorganic and organic colourants that are soluble or dispersed in the film (may or may not provide opacity).

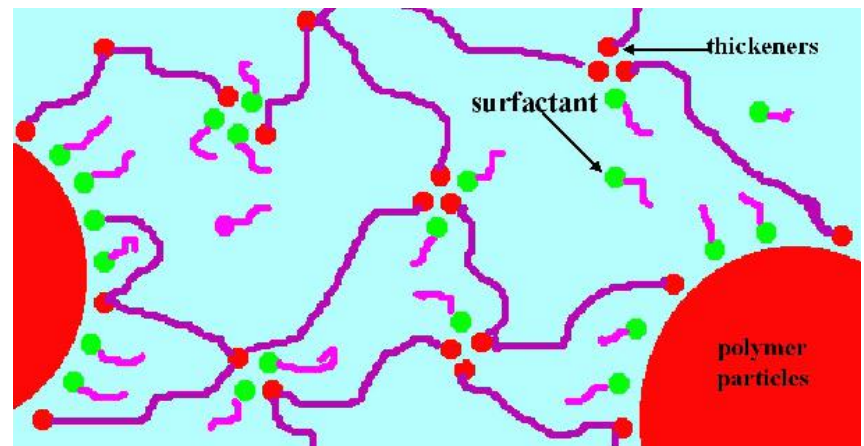


Thickeners

Thickeners are large water-soluble polymers added to a paint to increase its viscosity.

This property is important for a paint for several reasons:

- so the paint can flow out of the can
- so the paint can be applied to a substrate (glass, wood, steel, etc) using a paint brush or a roller.
- so the paint does not splatter or drip on the user
- so no brush marks can be seen
- to prevent settling of the paint in the can during storage
- so that a "good" film can be formed



Engineering Properties of Dried Film Coatings

Properties:

- Hardness
- Impact (& Chip) Resistance
- Flexibility
- Abrasion Resistance
- Solvent Resistance
- Adhesion

Tests:

- Indentation, Scratch (Pencil)
- Drop tests, Gravelometer
- Elongation, Bend
- Falling sand test
- MEK (methylethylketone) double rub
- Scraping, Crosscut Adhesion

Issues:

- Properties are a complex function of many factors: T_g , MW, crosslink density, pigmentation, static stress-strain behaviour, transient (creep) behavior
- What do you test? Free films? Coating with substrate?
- Weatherability and UV resistance – How does it hold up over time?

Methods for applying polymeric Coatings

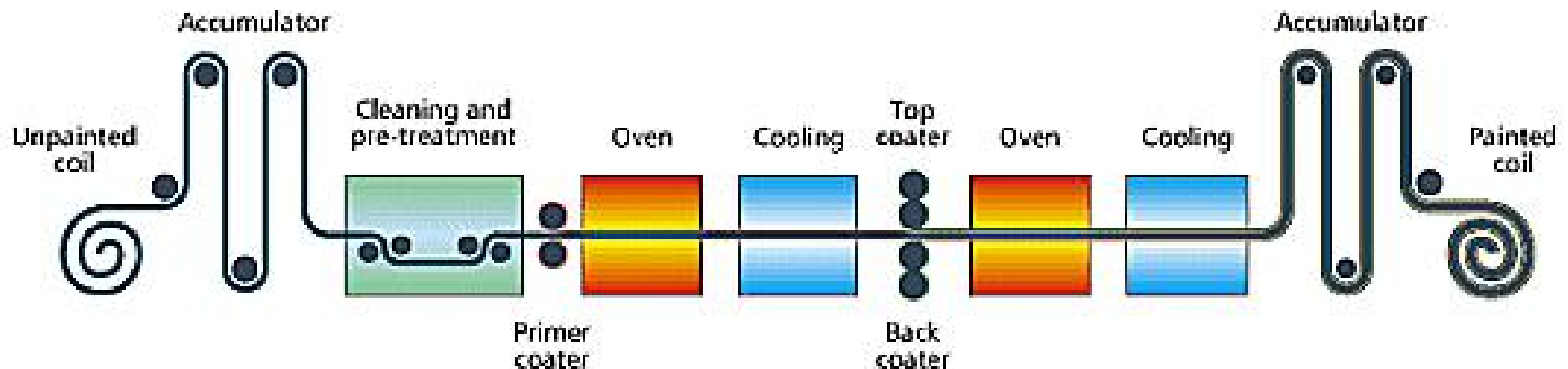
- Powder Spray coatings
- Electrocoating
- Fluidized Bed
- Dip coating
- Spray coating
- Spin coating
-



Coating processes: Coil Coating



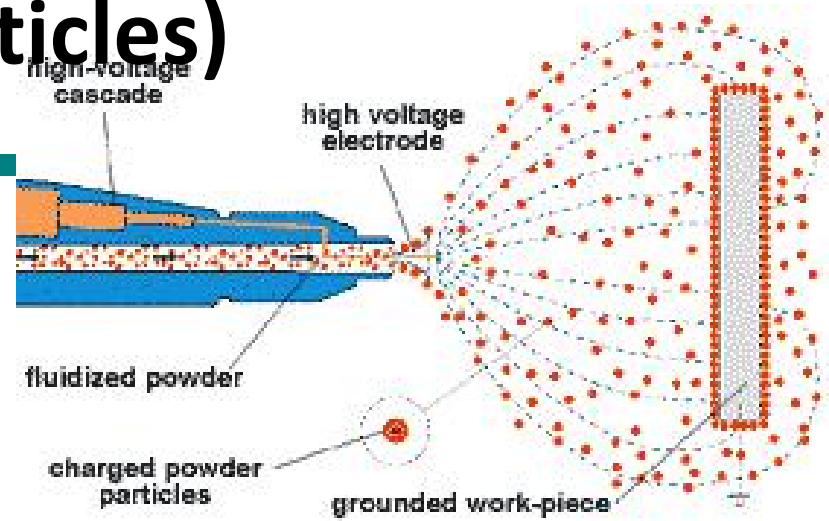
- Coat sheet metal from coils before shaping
- Calendar or knife delivery
- Also electrocoat & spray



Coil Coating



Spraying (charged particles)



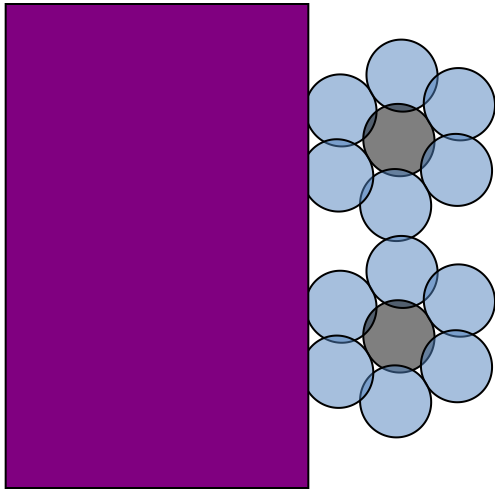
Curing
(Infrared oven)

Powder Coating

- High-gloss finish
- Uniform coat
- Durable coat
- Color selection
- Pollution-free
- Recycle powder
- Short cure times

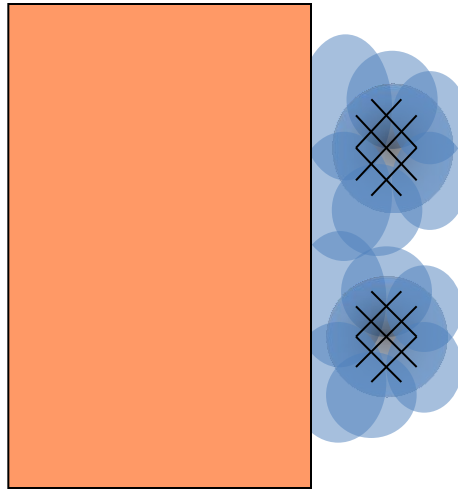


cold
substrate



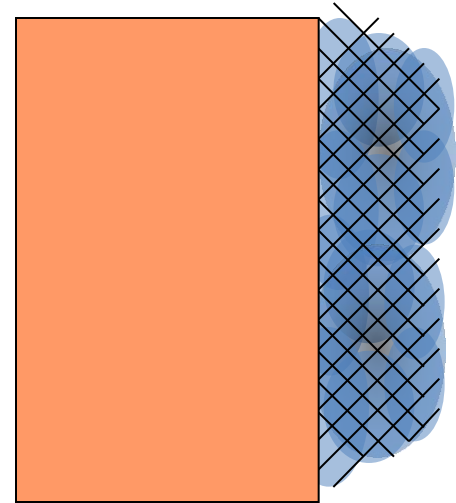
electrical
adhesion

substrate
heated



polyester melts
TGIC diffuses

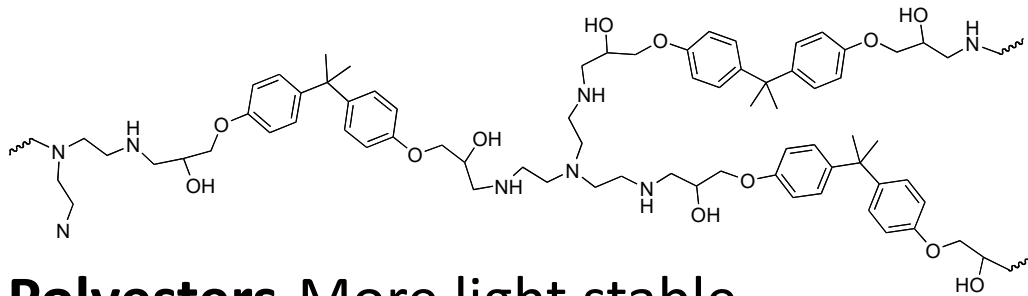
cross-
linked



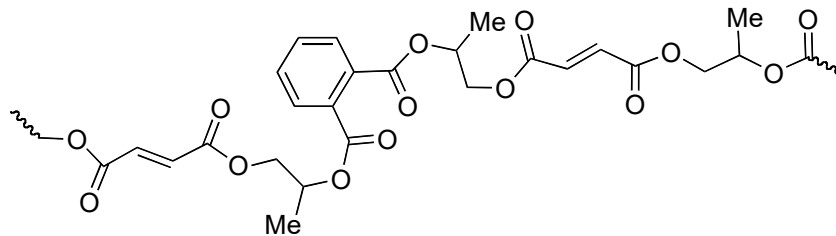
solid
film

Powders used in Spray Coating

Epoxies-More durable, but more light sensitive (chalking)



Polyesters-More light stable



Polymer
Binder

+

Pigment

+

Filler

+

Antioxidant

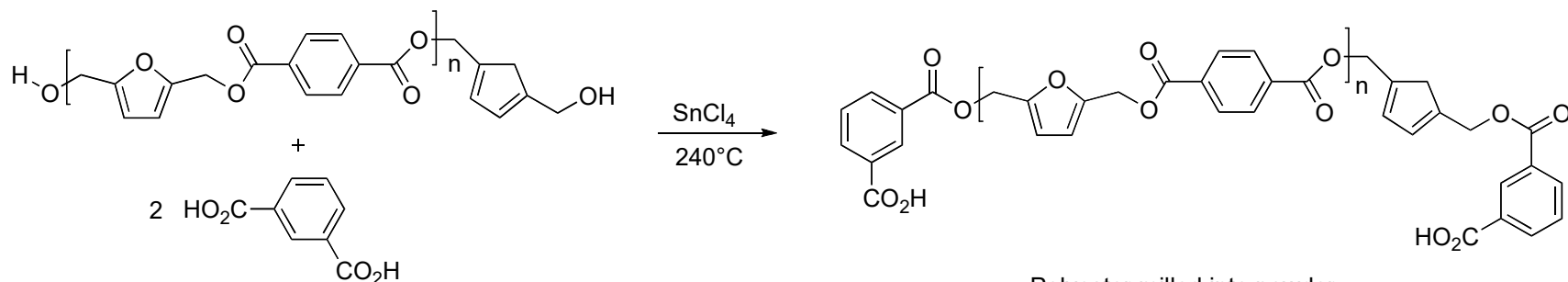
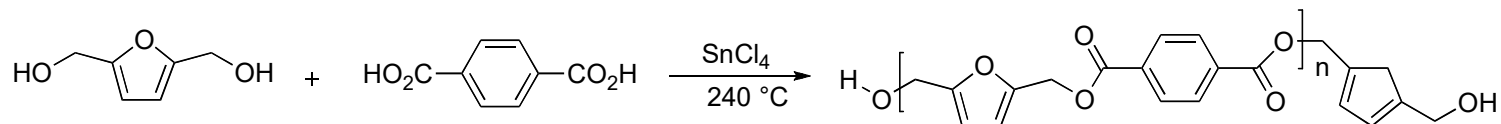
Applications: OEM

Following parts are powder coated with DuPont Powder Coatings:

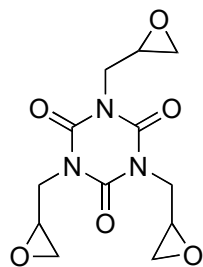


- Seats
- Radiators
- Battery boxes
- Brake parts
- Torsion bars
- Wheels
- Windscreen wipers
- Roof rails
- Oil filter boxes
- Magnesium doors
- Profiles for sunroof
- Springs
- Side board panels
- Car Bodies

Preparing the Coating Powder



Polyester milled into powder



tri-glycidyl-iso-cyanurate
powder

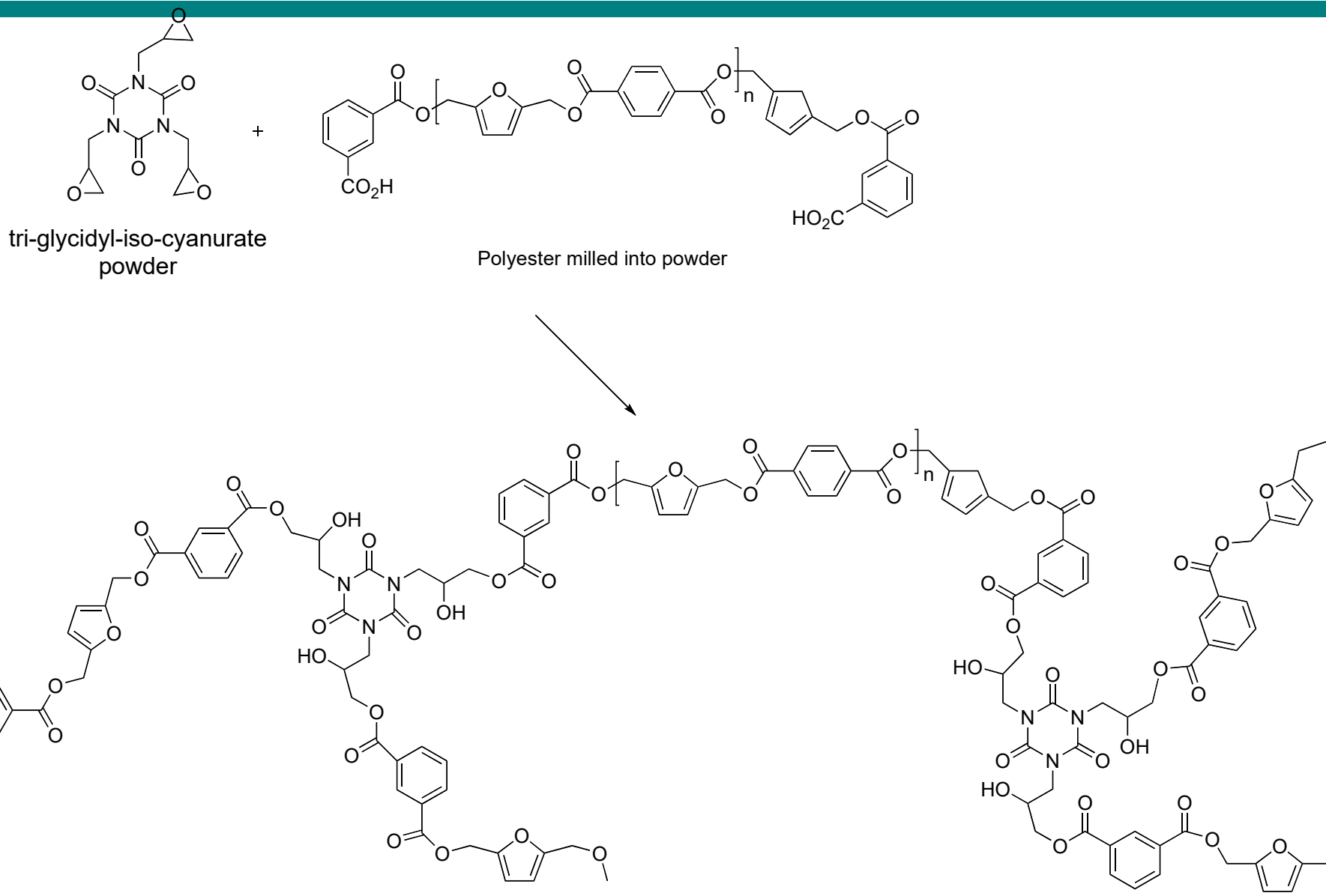
+

Polyester
powder

=

Paint

The crosslinking during the cure

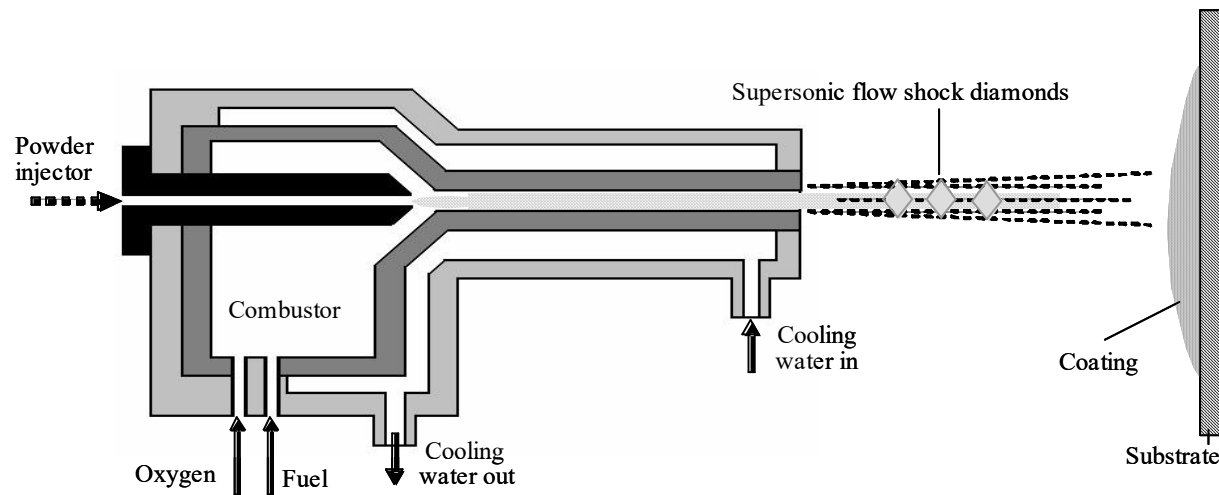
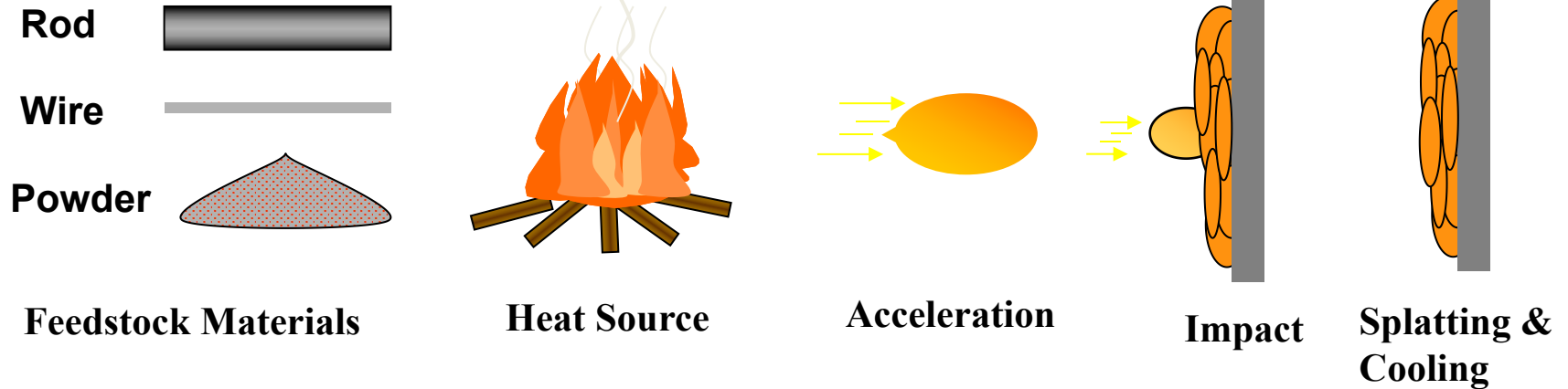


Powder Coating versus Paint

- no solvent - recycles unused coating - single layer sufficient	(toxic) solvent - overspray wasted - several layers required
conflicting flow requirements	viscosity adjusted separately

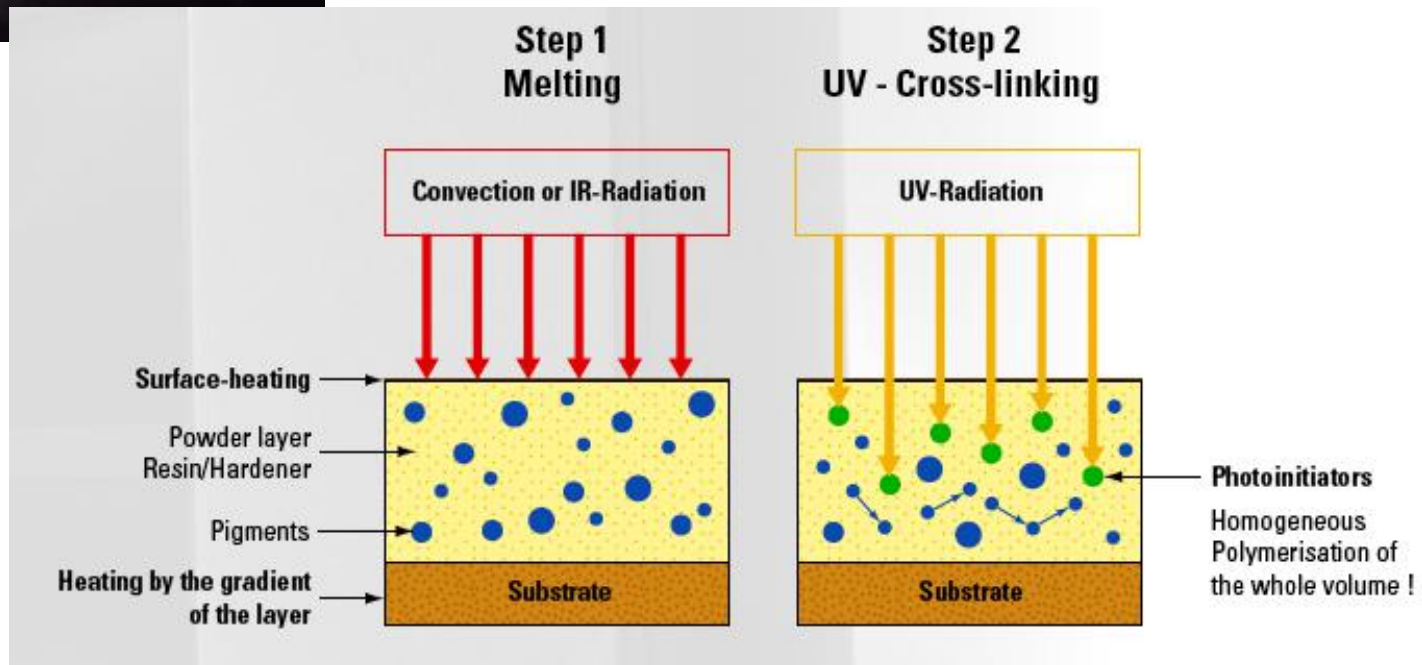
now dominates for coating of parts

Thermal Spray





Powder coating wood 90-120 ° C IR heating



DUPONT® STOCKS SEVERAL TYPES OF DECORATIVE POWDER COATINGS
THE FIRST LETTER: BINDER TYPE

- | | |
|--|---|
| E = Epoxy | U = Aliphatic-Urethane (called Urethane) |
| R = Epoxy-Polyester Hybrid | P = TGIC-Polyester |
| A = Aromatic-Urethane | S = Siloxane or Silicone Polyester |
| G = Polyester, TMMGU Glycouril curative | H = Non-TGIC Polyester, HAA Curative |

	Epoxy	Hybrid	Aromatic Urethane	Aliphatic Urethane	TGIC Polyester	Polyester Wrinkle	NON-TGIC Polyester
Hardness	●●●●●	●●●●●	●●●	●●●	●●●●●	●●●●●	●●●●●
Flexibility	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●
Overbake Stability	●●	●●●●●	●●	●●●●●	●●●●●	●●●	●●●●●
Exterior Durability	●	●	●●	●●●●●	●●●●●	●●●●●	●●●●●
Corrosion Protection	●●●●●	●●●●●	●●●	●●●	●●●●●	●●●	●●●
Chemical/Solvent Resistance	●●●●●	●●●	●●●	●●●	●●●●●	●●●	●●●
Ease of Application	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●	●●●●●

This table shows the difference among binder types in the various performance characteristics

● Not Recommended ●● Fair ●●● Good ●●●● Very Good ●●●●● Excellent

There are 4 key stages in the manufacture process of powder coatings:

1. Formulation

The formulation of powder coatings involves the use of 4 key raw material types: resins, pigments, curing agents and additives. Resins provide strength, durability and adhesion; pigments add color; curing agents ensure that the resins bond to the substrate and harden, and additives give further coating properties such as enhanced application, improved flow, etc.

2. Premix

Once a product has been formulated, raw materials are carefully weighed and measured. These raw materials are then blended together, either by mixing or tumbling, to ensure a consistent dispersion of raw materials throughout the mixture.

3. Extrusion

The mixed raw materials are then fed into an extruder. The extruder applies pressure and heat to melt the resins and thoroughly incorporate the pigments, curing agents and other ingredients into a homogenous substance. The extrudate is then cooled and chipped.

4. Grinding

Once the extrudate has been cooled and chipped, it is ready to be ground into a fine powder. This powder is then sieved and classified to ensure an even particle size distribution. Particles that are under 10 micrometers or over 120 micrometers can cause application problems and are restricted by the sieving and classification process.



Epoxy powder coated podium

Powder coatings have several significant advantages over solvent-based liquid paints:

- No solvents - Liquid paints can contain up to 70% solvent. Powder coatings contain no solvents, and therefore eliminate solvent release into the environment.
- Recyclable – Over sprayed powder can be reclaimed and reused, allowing up to 95% material utilization, reducing waste
- Fewer variables in film properties - The more uniform properties of powders and the elimination of thinners and solvent balancing lead to a more consistent paint film in just one coat.

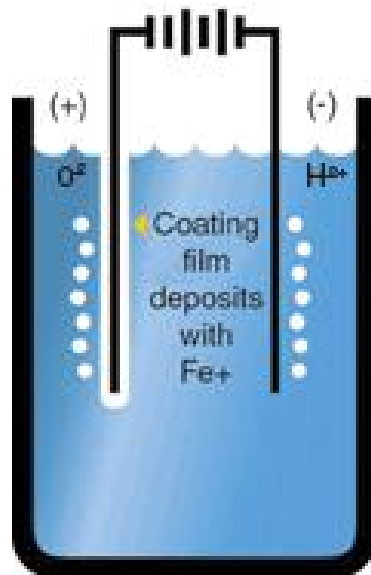
Electrocoating or E-coat

The Electrocoating Process...

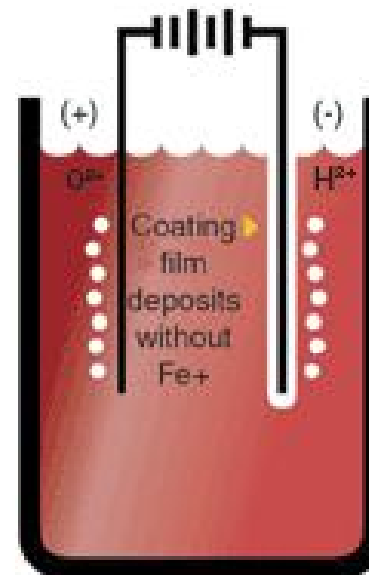
- Precipitation of paint particles onto a metal substrate
- Highly efficient and automated process
- Paint deposition is regulated by voltage
- Coating can be either anodic or cathodic
- Thermoset curing



Anodic Electrocoating



Cathodic Electrocoating



E-Coat: Anodic Coatings

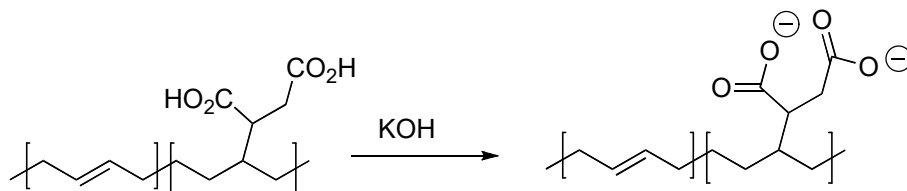
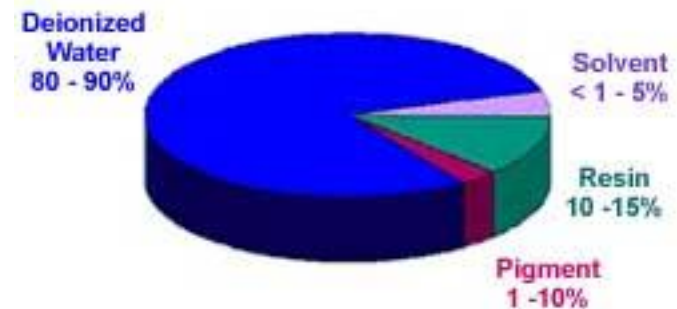
Anode has a positive charge that attracts the negatively charged polymers

Epoxies (cure $> 80^{\circ}\text{C}$)

Acrylics (cure $> 150^{\circ}\text{C}$)

Acid-modified polybutadiene

Butylated-formaldehyde-melamine (150°C)

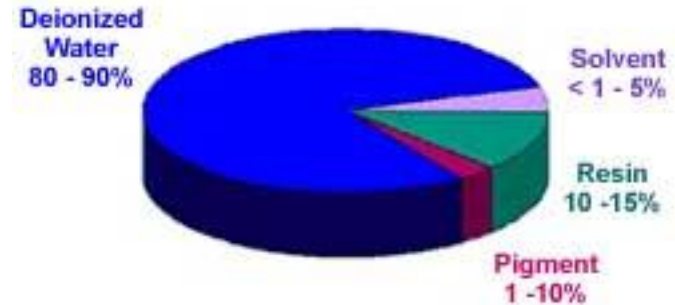


E-Coat: Cathodic Coatings

Cathode has a negative charge that attracts the positively charged polymers

Epoxies (cure $>190^{\circ}\text{C}$)

Acrylics (cure $>190^{\circ}\text{C}$)

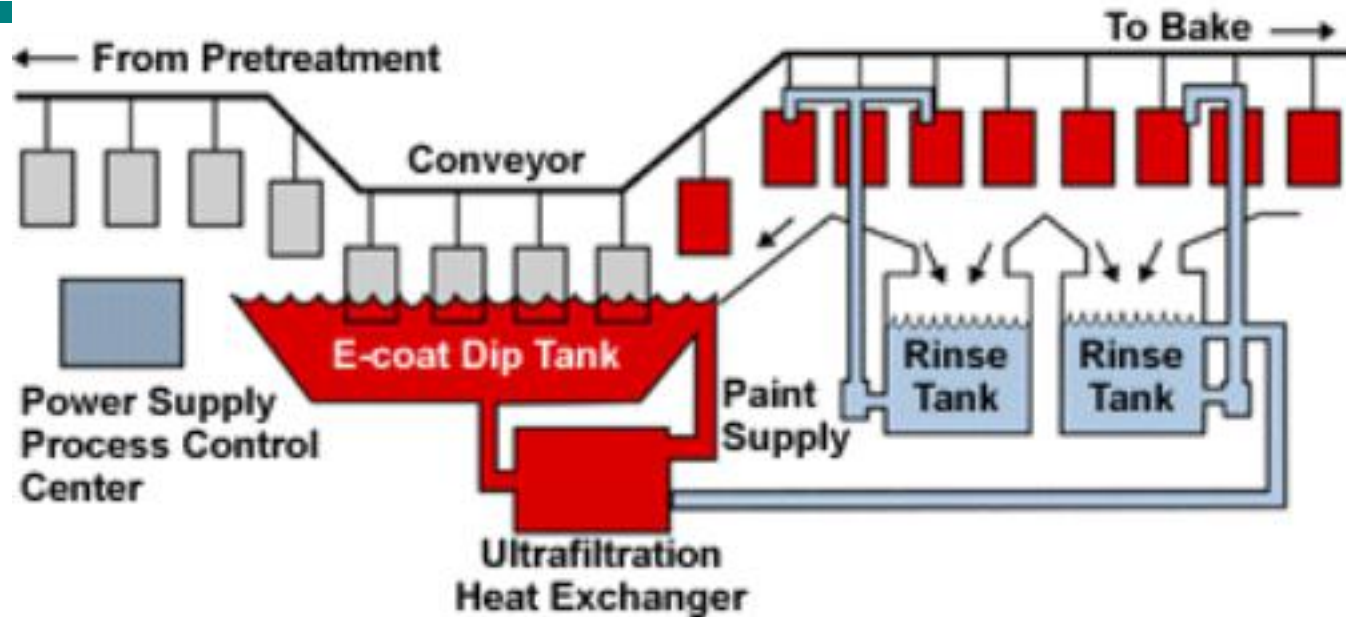


- Corrosion resistant cationic epoxies
- High edge coverage for sharp objects
- Solvent free anodic products
- Cathodic acrylics with corrosion resistance and excellent exterior durability
- Near 100% transfer efficiency operation
- Two-coat Electrocoat for severe environments
- Heavy-Metal free formulations
- Electrodepositable Photoresists

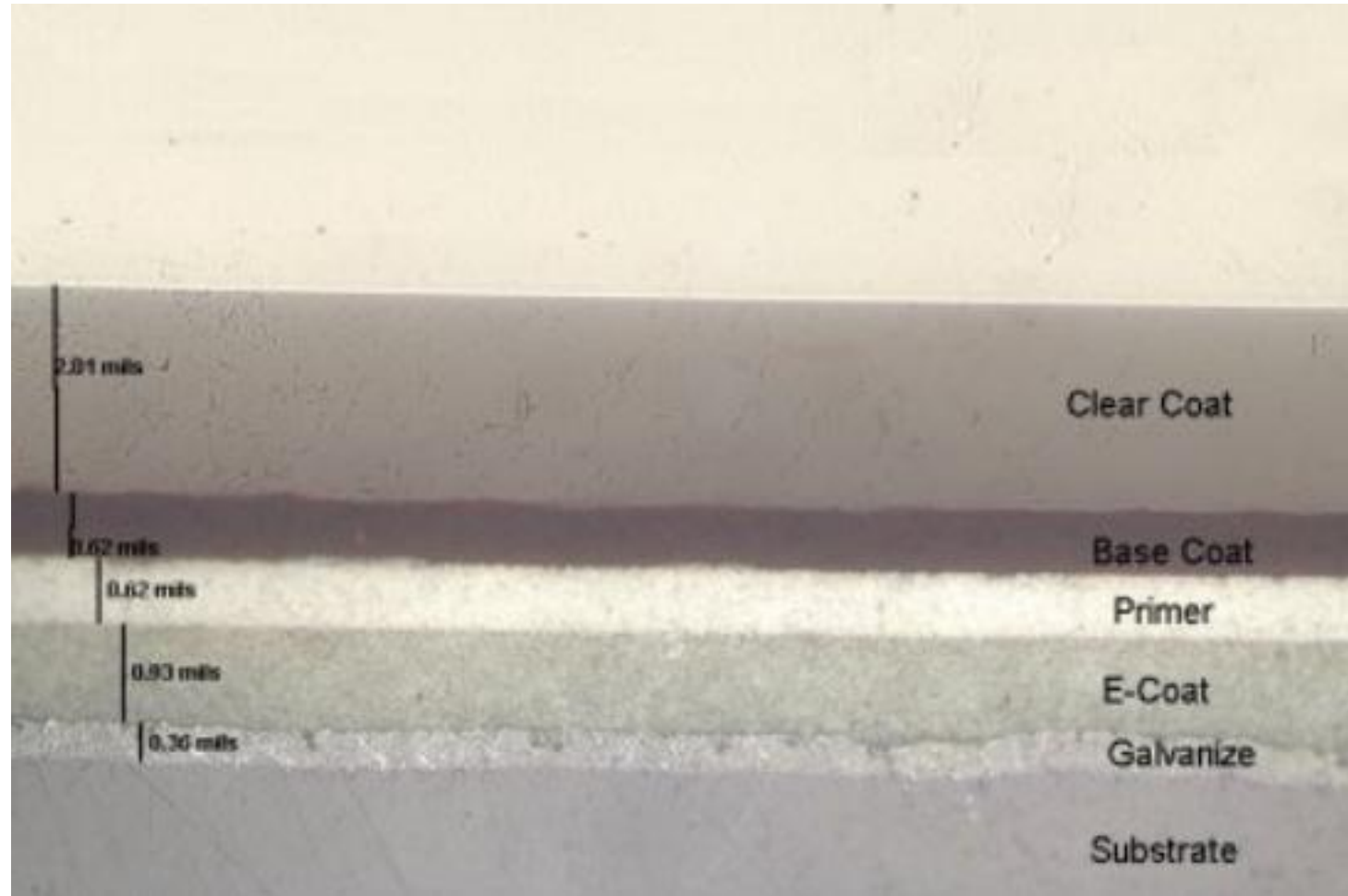


Agriculture Equipment, Appliance,
Automotive, Fasteners, Metal Office
Furniture, Printed Circuit Boards, Structural
Steel, Wheels, Railway

Typical Electrocoat System

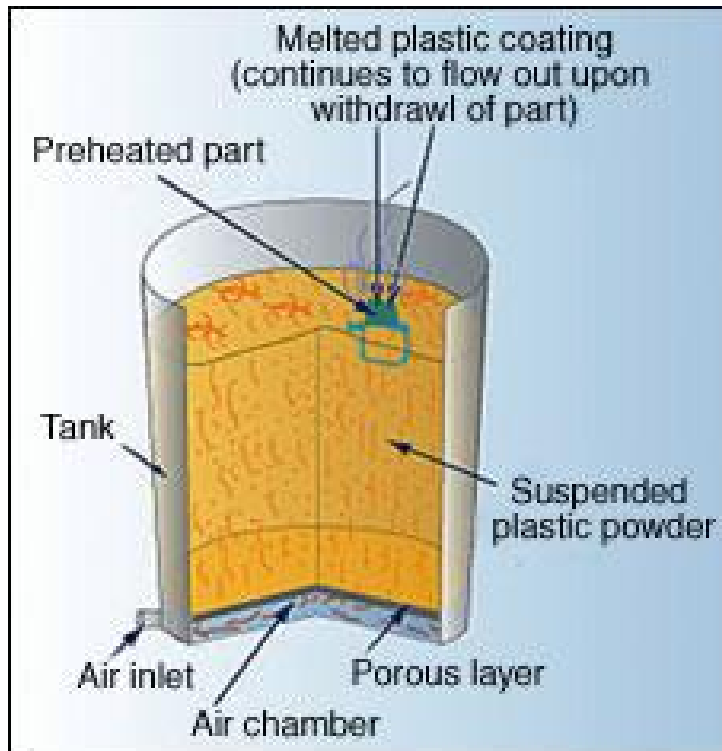
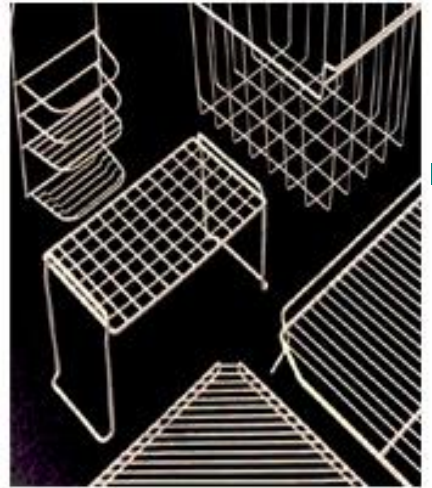


Automobile Painting



Fluidized Bed Coating

- Polymer powder in air generated fluidized bed
- Normal or electrostatic FB coater
- PVC, polyethylene copolymers
- Protective, decorative coatings
- Safety glass bottles



Heat part, then dip into FB coater

100 ° C for PE copolymer

Fluidized Bed Coatings



A part recently taken out of a fluid bed coater with about 10 mils of functionalized polyethylene applied.

Dip Coating



PVC

CVD Polymerization

- Parylene is the generic name for poly-para-xylylene, a completely linear, highly crystalline material
- Vapor Deposition in Vacuum (conformal coating)
- Excellent Dielectric (> 5000 V/mil)
- Excellent Strength (Yield & Tensile strength > 8000 psi)
- Highly stable (Insoluble in most Solvents)
- Biocompatible
- Water absorption ($< 0.1\%$ in 24hrs)

CVD: Parylene Coating



*Conventional Coating
(uneven)*

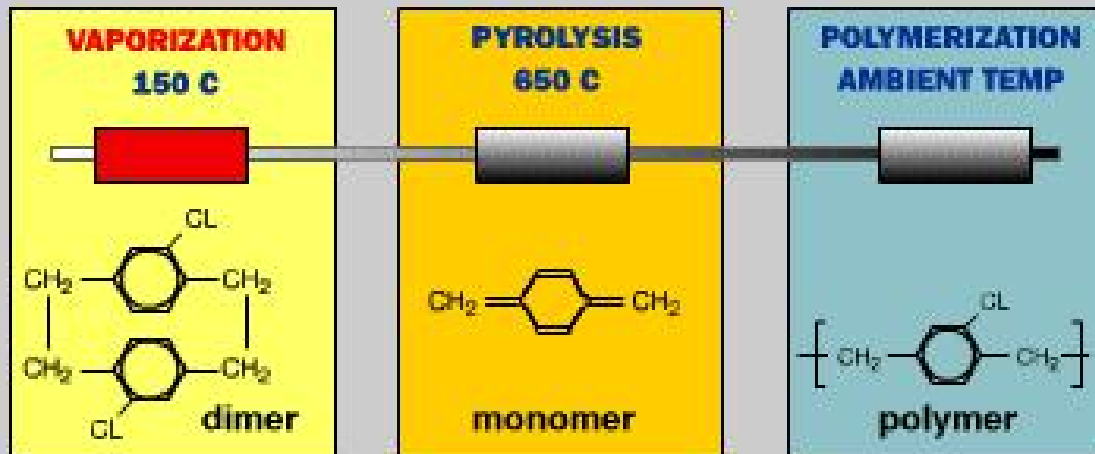


*Parylene Coating
(uniform)*



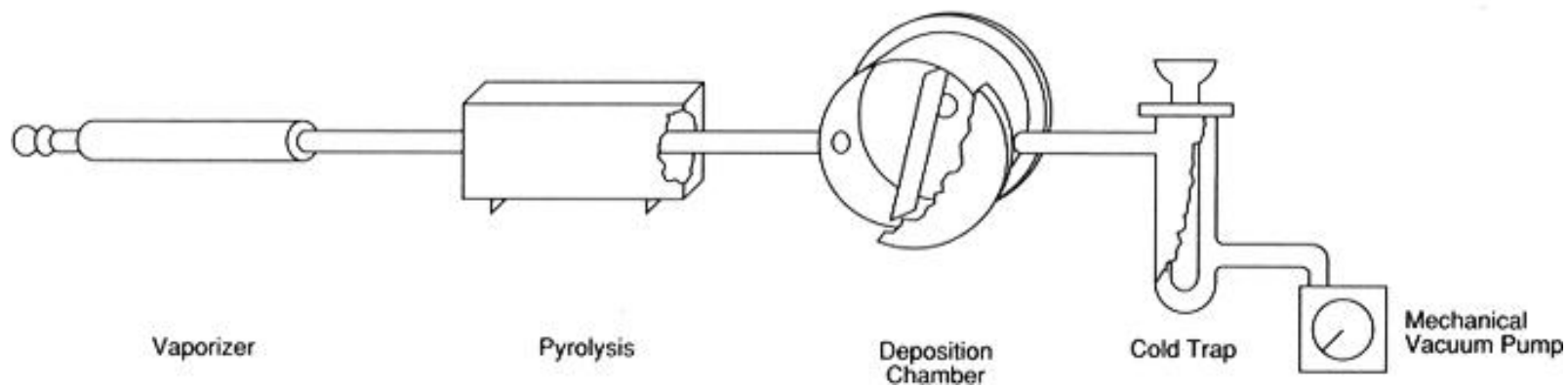
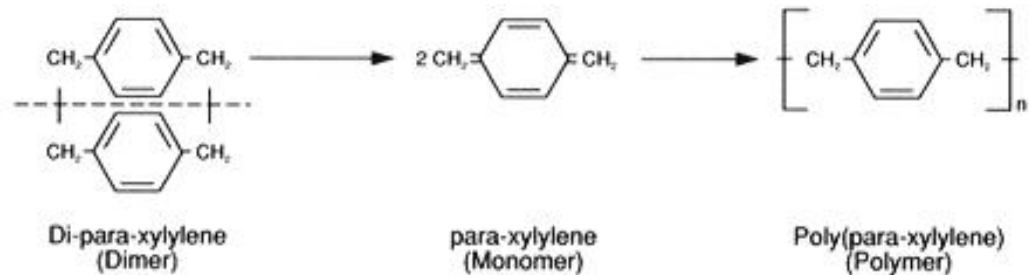
CVD Polymerization

- Gas phase
- Polymerizes on contact
- Conformal coatings
- Pinhole free
- Preserving artifacts (paper)
- Microelectronics
- Medical devices



In the **Vaporization Zone** Parylene dimer vaporizes under vacuum at 150-170 degrees Celcius.

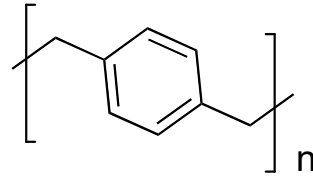




Polyxylylene

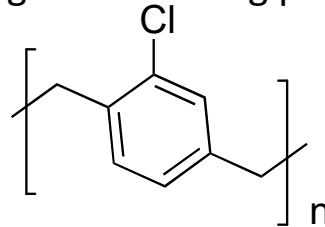
Parylene C

The most widely used dimer, providing a useful combination of properties, plus a very low permeability to moisture, chemicals, and other corrosive gases.



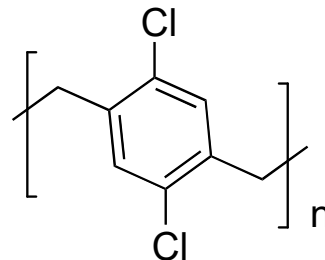
Parylene N

Provides high dielectric strength and a dielectric constant that does not vary with changes in frequency. Best selection where greater coating protection is required.



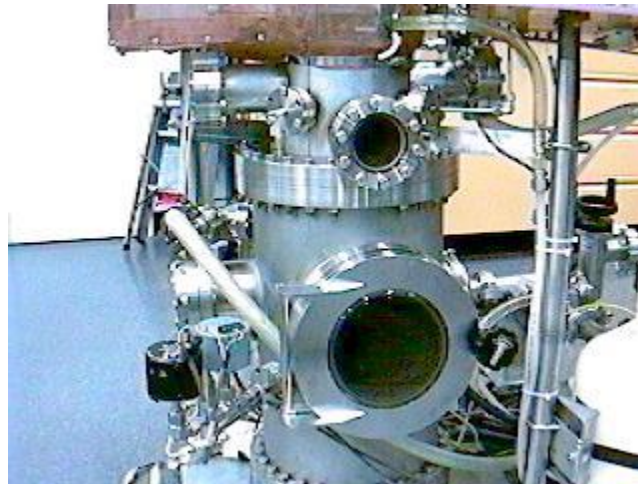
Parylene D

Maintains its physical strength and electrical properties at higher temperature

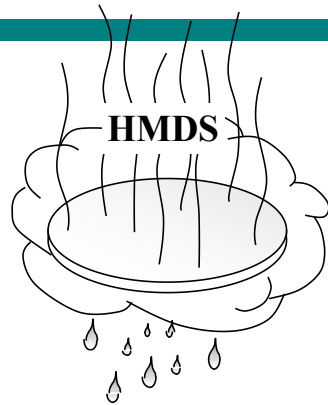


RF Plasma Source

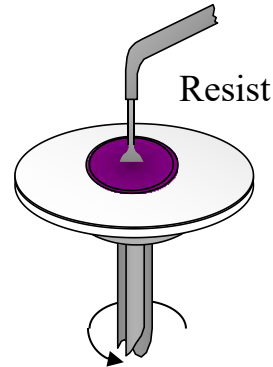
- Capacitive Coupling
- Inductive Coupling



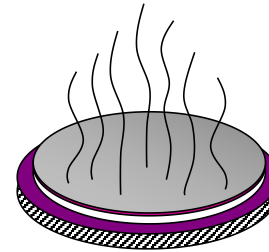
Eight Steps of Photolithography



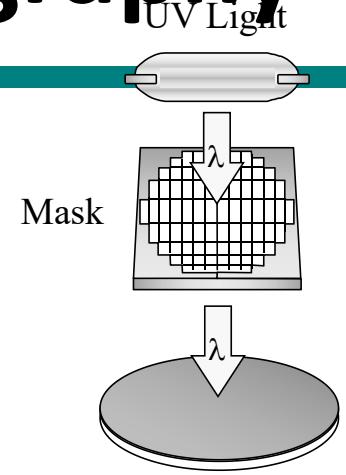
1) Vapor prime



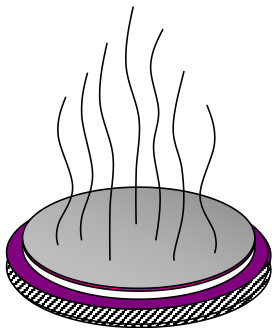
2) Spin coat



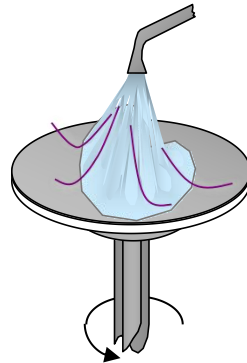
3) Soft bake



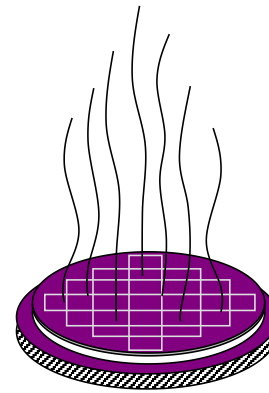
4) Alignment and Exposure



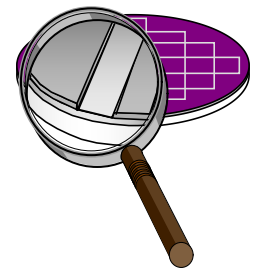
5) Post-exposure bake



6) Develop



7) Hard bake

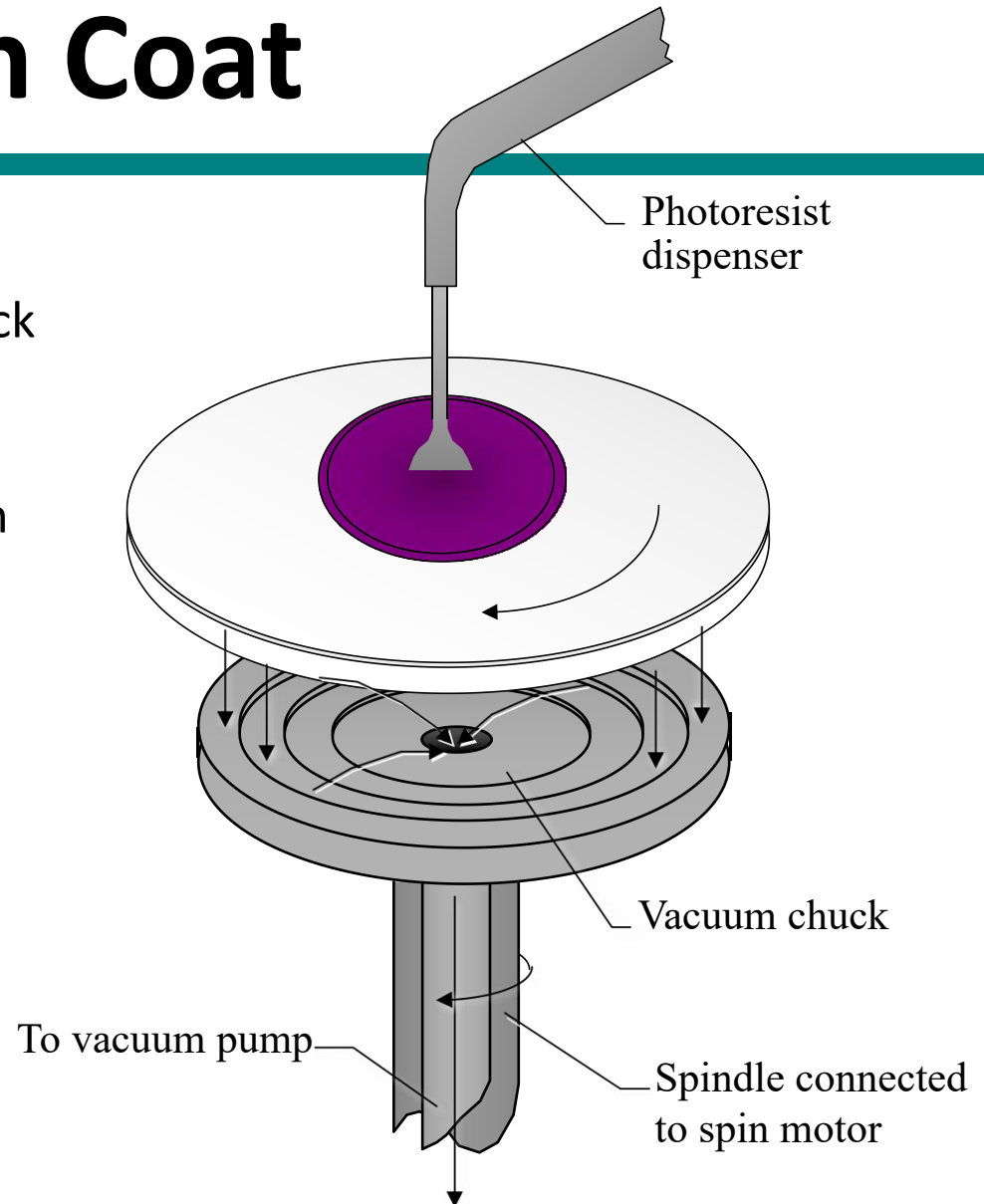


8) Develop inspect

Spin Coat

Process Summary:

- Wafer is held onto vacuum chuck
- Dispense ~5ml of photoresist
- Slow spin ~ 500 rpm
- Ramp up to ~ 3000 to 5000 rpm
- Quality measures:
 - time
 - speed
 - thickness
 - uniformity
 - particles and defects



HOMEWORK

- Introduce at least 2 WATERPROOF polymer coatings and explain their principles
- Encourage you to give a short talk on what you find during next class.